From the Cubic Anomaly to the Standard Model

A Uniqueness and No-Go Theorem for Compact Homogeneous Chiral Matter

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Abstract

Conventional unification usually chooses an internal symmetry, a matter representation, and a family multiplicity as separate inputs. We instead classify the full domain of connected, infinitesimally effective compact pairs $H \subsetneq G$ for which the canonical isotropy module $\mathfrak{q} = \mathfrak{g}/\mathfrak{h}$ is irreducible of complex type and the cubic anomaly of one complex half has a nonzero radical. Up to isomorphism of Lie-algebra pairs and duality, the classification has one solution:

$$\mathfrak{e}_8 \supset \mathfrak{e}_6 \oplus \mathfrak{su}(3)_M, \qquad V = \mathbf{27} \boxtimes \mathbf{3}, \qquad \mathrm{Rad}(c_V) = \mathfrak{e}_6.$$

Thus the ambient algebra, complex matter module, cubic-anomaly-safe ideal, and three-dimensional multiplicity factor are fixed simultaneously. The census is direct: full-rank pairs are enumerated by toral gradings, while rank-deficient exceptional and classical pairs are excluded by index arithmetic and uniform highest-weight arguments.

For the selected E_6, we exhaust the $27 \cdot 36 \cdot 40 = 38{,}880$ triples formed from the three Weyl orbits of maximal proper closed root subsystems. The triples form six Weyl orbits; exactly two have a common A_2 component. Their faithful connected groups give

$$G_{\mathrm{SM}} \subset G_{\mathrm{LR}} \subset G_{\mathrm{PS}} \subset \mathrm{Spin}(10) \subset E_6^{\mathrm{sc}}, \qquad Y = T_R^3 + \frac{B-L}{2},$$

with exact global quotients and compatible charge lattices.

Restricting the selected module to G_{SM} gives

$$V|_{G_{\mathrm{SM}}} \cong 3\,\mathcal{F}_{\mathrm{SM}} \oplus 3\,(\mathbf{5} \oplus \overline{\mathbf{5}})|_{G_{\mathrm{SM}}} \oplus 6\,\mathbf{1}.$$

Here $\mathcal{F}_{\mathrm{SM}}$ denotes one Standard Model family. The two complex orientations have chiral indices $+3$ and -3; all remaining summands are vectorlike or singlets. Under the complete-matter interpretation (M), the family multiplicity is three. Structures normally entered independently in grand-unified model building are here components of one classified homogeneous structure.

1 Introduction

The inverse problem. A chiral gauge theory normally begins with three choices: an internal symmetry algebra, a representation for left-handed Weyl fermions, and a replication number. The Standard Model spectrum is chiral and anomaly-free, contains an unbroken $SU(3)$ color factor, and occurs in three families [1]. Minimal $SU(5)$, Spin(10), and E_6 unification organize one family in $\mathbf{10} \oplus \overline{\mathbf{5}}$, $\mathbf{16}$, or $\mathbf{27}$, but leave the family number free [20, 21, 22]. This paper asks the converse question: can the group, its matter, and the multiplicity be outputs of one classification rather than independent inputs?

For a connected compact pair $H \subset G$, the quotient

$$\mathfrak{q} = \mathfrak{g}/\mathfrak{h}$$

is a canonical H-module. The embedding therefore supplies an internal algebra and a matter representation at once. We range over every connected, infinitesimally effective compact pair for which this module is irreducible of complex type and its cubic anomaly has a nonzero ideal-valued radical. No exceptional group, rank, representation dimension, family number, or Standard Model subgroup is fixed in advance.

Why this domain. The defining conditions have independent roles. Compactness is the standard finite-dimensional setting for unitary internal symmetries. Connectedness isolates the continuous symmetry seen by Lie theory; discrete extensions can be studied after the connected classification. Infinitesimal effectivity removes generators that act on nothing. Complex type is the representation-theoretic condition that distinguishes a module from its conjugate; choosing one of its two orientations gives a chiral half. Irreducibility states the strict homogeneous unity hypothesis: the matter complement is one structure, not a hand-assembled direct sum. Finally, the anomaly radical asks the representation itself to identify a nonzero ideal free of every mixed cubic insertion. It does not choose a cubic-anomaly-safe subgroup after the fact. Together these conditions define a physically motivated and answer-neutral unification domain.

The homogeneous premise itself has a long pedigree. Isotropy modules organize fields in coset-space dimensional reduction, compact internal geometry can produce chiral fermions, and exceptional Kähler cosets have been used for family unification [2, 3, 4, 5, 6, 7, 8]. Heterotic standard embeddings likewise express net chirality through chosen compactification data [9], and recent bordism work sharpens the anomaly analysis of exceptional family-unification cosets [26]. These approaches choose a particular group, coset, bundle, or geometry and compute its consequences. The new step here is the inverse classification: no pair or matter representation is selected before the census.

Anomaly cancellation also has a history as a source of structure rather than a final check. In the early electroweak theory it linked quark and lepton content and supplied an independent consistency argument for a fourth quark [11, 12]. The present theorem applies that logic one level earlier, before the internal algebra and its matter module have been chosen.

What is new. Several ingredients of the final chain are classical. The branchings $E_8 \to E_6 \times SU(3)$ and $E_6 \to \mathrm{Spin}(10)$, and the faithful Standard Model group as the intersection of $SU(5)$ and Pati–Salam inside Spin(10), are familiar [23, 24]; exceptional subgroup intersections also appear in Jordan-algebra constructions [25]. The new content is the converse and its simultaneity:

1. the mixed anomaly radical leaves one homogeneous pair and excludes every other full-rank and rank-deficient possibility in the domain;

2. the complete 38,880-triple E_6 space has six Weyl orbits, exactly two with a common A_2 component;

3. those two orbits determine the faithful Standard Model and left–right chains, their compatible finite kernels, and 8,640 labelled containment incidences representing 2,160 distinct root-subsystem inclusions; and

4. the selected module has chiral index of magnitude three.

The familiar embeddings are therefore identified outputs, not assumptions of the argument.

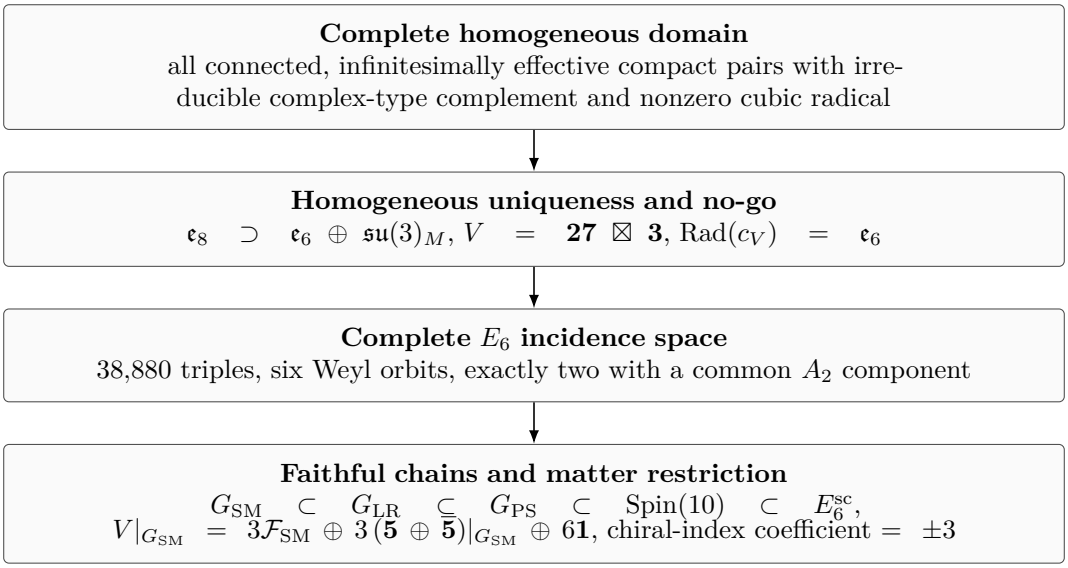


Figure 1: The logical architecture. The first arrow is the homogeneous classification, the second the exhaustive E_6 orbit classification, and the third applies the pair-first and A_2-extremal criteria before the branching and lattice calculation. Interpretation (M) enters only when the final chiral index is identified with the complete physical family count.

How to read the paper. Section 2 defines the two invariant conditions and states the unique-pair theorem. Section 3 proves the exhaustive homogeneous census. Section 4 classifies the finite E_6 triple space and reconstructs the faithful subgroup chains. Section 5 restricts the selected module and computes its chiral index and anomalies. Readers interested first in the result can read Theorem 2.1, Theorems 4.2 and 4.3, Corollary 4.7, and Section 5; the intervening sections and appendices supply the full proofs and independent exact verification.

2 Homogeneous chirality and cubic anomalies

This section turns the physical vocabulary into two invariant algebraic conditions. The first produces one irreducible real module with two complex orientations. The second asks its cubic anomaly to identify a nonzero ideal that is safe against every mixed insertion.

Let G be a connected compact Lie group and let $H \subsetneq G$ be a connected closed subgroup. Write $\mathfrak{q} = \mathfrak{g}/\mathfrak{h}$ for the isotropy representation and assume infinitesimal effectivity: $\mathfrak{h}$ contains no nonzero ideal of $\mathfrak{g}$. An $\mathrm{Ad}(G)$-invariant inner product gives

$$\mathfrak{q} = \mathfrak{g}/\mathfrak{h} \simeq \mathfrak{h}^{\perp}, \qquad \mathfrak{g} = \mathfrak{h} \oplus \mathfrak{q}, \qquad [\mathfrak{h}, \mathfrak{q}] \subseteq \mathfrak{q}.$$

Compactness makes the isotropy representation completely reducible. The pair itself fixes the underlying module as $\mathfrak{g}/\mathfrak{h}$.

Fraktur letters denote Lie algebras. For a representation R, R^* denotes its dual and $\overline{R}$ its complex conjugate; for compact groups these are isomorphic. We use R^* in duality statements and bars in conventional representation labels. The symbol $\boxtimes$ denotes an external tensor product, and a superscript "sc" denotes the simply connected compact group.

2.1 Complex-type isotropy

The algebra $\mathrm{End}_{\mathfrak{h}}(\mathfrak{q})$ consists of the real linear maps on $\mathfrak{q}$ that commute with $\mathfrak{h}$. We require

$$\boxed{\mathrm{End}_{\mathfrak{h}}(\mathfrak{q}) \cong \mathbb{C}}\,. \tag{1}$$

Compactness makes $\mathfrak{q}$ completely reducible. If it had a proper invariant summand, projection onto that summand would be a nontrivial idempotent in $\mathrm{End}_{\mathfrak{h}}(\mathfrak{q})$. Since $\mathbb{C}$ has no such idempotent, $\mathfrak{q}$ is irreducible. Choose J corresponding to $i \in \mathbb{C}$. Then $J^2 = -I$, and J is unique up to sign. Its $+i$- and $-i$-eigenspaces in $\mathfrak{q}_{\mathbb{C}}$ are irreducible dual $\mathfrak{h}$-modules V and V^*, so

$$\mathfrak{q}_{\mathbb{C}} = V \oplus V^*, \qquad V \ncong V^*.$$

The two choices J and $-J$ exchange V and V^*. The homogeneous data determine the unordered dual pair $\{V, V^*\}$, but not an orientation. Choosing one member gives the usual all-left-handed convention; choosing the other reverses all charges.

2.2 The cubic anomaly and its radical

Because H is compact, choose an H-invariant Hermitian inner product on V. The operators $\rho_V(X)$ are then skew-Hermitian, so $T_X = -i\rho_V(X)$ is Hermitian. With $\{A, B\} = AB + BA$, define

$$\begin{aligned} c_V(X,Y,Z) &= \frac{1}{2}\,\mathrm{Tr}_V(T_X\{T_Y, T_Z\}), \\ \Phi_V(X) &= c_V(X,\cdot,\cdot), \qquad\qquad \mathrm{Rad}(c_V) = \ker \Phi_V. \end{aligned} \tag{2}$$

For a left-handed Weyl multiplet, c_V is the group factor in the local triangle anomaly [10]. The map $\Phi_V : \mathfrak{h} \to \mathrm{Sym}^2\,\mathfrak{h}^*$ is equivariant, so its kernel is invariant under the adjoint action and is therefore an ideal. It is the largest ideal $\mathfrak{a} \lhd \mathfrak{h}$ for which every cubic with one argument in $\mathfrak{a}$ vanishes. Equivalently,

$$\exists\, 0 \neq \mathfrak{k} \lhd \mathfrak{h} :\ c_V(\mathfrak{k},\mathfrak{h},\mathfrak{h}) = 0 \quad \Longleftrightarrow \quad \mathrm{Rad}(c_V) \neq 0.$$

We require

$$\mathrm{Rad}(c_V) \neq 0. \tag{3}$$

This is the second homogeneous condition. For a particular nonzero ideal $\mathfrak{k}$, it is stronger than vanishing only on $\mathfrak{k}^3$:

$$\begin{aligned} \text{vanishing only on } \mathfrak{k}^3 \quad &: \quad c_V(\mathfrak{k},\mathfrak{k},\mathfrak{k}) = 0, \\ \text{vanishing with one } \mathfrak{k}\text{-argument} \quad &: \quad c_V(\mathfrak{k},\mathfrak{h},\mathfrak{h}) = 0. \end{aligned} \tag{4}$$

Every such ideal lies in $\mathrm{Rad}(c_V)$. Hence every component with one argument in the radical vanishes, including components with arbitrary $\mathfrak{h}$-valued background insertions. Local counterterms can redistribute the consistent anomaly among currents, but they do not alter the invariant polynomial represented by c_V [10]. Condition (3) applies to that invariant polynomial, not merely to a chosen subalgebra. Duality sends c_V to $-c_V$ and leaves its radical unchanged.

Theorem 2.1 (Uniqueness of the homogeneous pair)**.** *Among the connected, infinitesimally effective compact pairs specified above, conditions* (1) *and* (3) *have, up to isomorphism of Lie-algebra pairs and exchange of V with V^*, the unique solution*

$$(\mathfrak{g}, \mathfrak{h}, \mathrm{Rad}(c_V), \{V, V^*\}) = (\mathfrak{e}_8, \mathfrak{e}_6 \oplus \mathfrak{su}(3)_M, \mathfrak{e}_6, \{\mathbf{27} \boxtimes \mathbf{3}, \overline{\mathbf{27}} \boxtimes \overline{\mathbf{3}}\}). \tag{5}$$

Conversely, this pair satisfies the hypotheses.

Remark (Contrapositive and no-go consequence)*.* Every other connected, infinitesimally effective compact pair with irreducible complex-type isotropy has $\mathrm{Rad}(c_V) = 0$. No candidate in the full census has $\mathrm{Rad}(c_V) = \mathfrak{h}$; the survivor is the unique case with a proper nonzero radical. Equivalently, any different homogeneous realization must fail at least one pair hypothesis or one of conditions (1) and (3). Group-level global forms are determined separately in Section 4.

Remark (What the theorem fixes)*.* The pair fixes $\mathfrak{h}$ and $\mathfrak{q}$ simultaneously. In the surviving case the cubic radical then separates $\mathfrak{h} = \mathfrak{e}_6 \oplus \mathfrak{su}(3)_M$ into the cubic-anomaly-safe ideal $\mathfrak{e}_6$ and the commuting multiplicity factor $\mathfrak{su}(3)_M$. These are outputs of one theorem rather than separate group, representation, and family-symmetry choices.

Remark (Invariant degrees and the surviving radical)*.* The basic invariant degrees of $W(E_6)$ are $2, 5, 6, 8, 9, 12$. The cubic trace polynomial of any E_6-representation therefore vanishes on a Cartan subalgebra, and cubic gauge anomalies remain zero after restriction to subgroups of E_6. By contrast, $W(A_2)$ has invariant degrees $2, 3$. For $V = \mathbf{27} \boxtimes \mathbf{3}$, the nonzero cubic lies in the $SU(3)_M$ direction, while every component with an E_6 insertion vanishes. This difference in invariant degrees accounts for $\mathrm{Rad}(c_V) = \mathfrak{e}_6$.

Lemma 2.2 (Structural normal form of the homogeneous domain)**.** *Under the standing pair hypotheses and Conditions* (1) *and* (3)*, $\mathfrak{q}$ is an irreducible real $\mathfrak{h}$-module, the isotropy action is faithful, $\mathfrak{g}$ is simple, and $\mathfrak{h}$ is semisimple and maximal among proper Lie subalgebras of $\mathfrak{g}$.*

Proof. Maximality. Irreducibility was proved after (1). An intermediate algebra $\mathfrak{h} \subsetneq \mathfrak{l} \subsetneq \mathfrak{g}$ would then make $\mathfrak{l}/\mathfrak{h}$ a proper nonzero $\mathfrak{h}$-submodule of $\mathfrak{q}$. Hence $\mathfrak{h}$ is maximal among proper subalgebras of $\mathfrak{g}$.

Faithfulness. Let $\mathfrak{k}_0$ be the kernel of the isotropy representation. Since $\mathfrak{k}_0$ acts trivially on $\mathfrak{g}/\mathfrak{h}$, $[\mathfrak{k}_0, \mathfrak{q}] \subseteq \mathfrak{h}$. But $[\mathfrak{h}, \mathfrak{q}] \subseteq \mathfrak{q}$, so this bracket lies in both summands and vanishes. Since $\mathfrak{k}_0 \lhd \mathfrak{h}$, this makes $\mathfrak{k}_0$ an ideal of $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{q}$ contained in $\mathfrak{h}$. Infinitesimal effectivity therefore gives $\mathfrak{k}_0 = 0$. Because an element acting trivially on V also acts trivially on V^*, the $\mathfrak{h}$-module V is faithful.

Simplicity of $\mathfrak{g}$. The projection of $\mathfrak{z}(\mathfrak{g})$ to $\mathfrak{q}$ is fixed by $\mathfrak{h}$. If it were nonzero, irreducibility would make all of $\mathfrak{q}$ fixed. Every subspace would then be invariant, forcing $\dim \mathfrak{q} = 1$; its endomorphism algebra would be $\mathbb{R}$, contrary to (1). It follows that $\mathfrak{z}(\mathfrak{g}) \subseteq \mathfrak{h}$. It is an ideal of $\mathfrak{g}$, so infinitesimal effectivity gives $\mathfrak{z}(\mathfrak{g}) = 0$; compactness makes $\mathfrak{g}$ semisimple.

Suppose $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$, where $\mathfrak{g}_1$ is simple and $\mathfrak{g}_2 \neq 0$ is the sum of the remaining simple ideals. Infinitesimal effectivity implies that neither ideal is contained in $\mathfrak{h}$. Since $\mathfrak{h}$ is maximal and each $\mathfrak{g}_i$ is an ideal,

$$\mathfrak{h} + \mathfrak{g}_1 = \mathfrak{h} + \mathfrak{g}_2 = \mathfrak{g}.$$

If $X \in \mathfrak{h} \cap \mathfrak{g}_1$, write any $Y \in \mathfrak{g}$ as $Y = Y_\mathfrak{h} + Y_2$, with $Y_\mathfrak{h} \in \mathfrak{h}$ and $Y_2 \in \mathfrak{g}_2$. Then $[X, Y] = [X, Y_\mathfrak{h}] \in \mathfrak{h}$, so X acts trivially on $\mathfrak{g}/\mathfrak{h}$. Faithfulness gives $\mathfrak{h} \cap \mathfrak{g}_1 = 0$, and similarly $\mathfrak{h} \cap \mathfrak{g}_2 = 0$. If $p_i : \mathfrak{g} \to \mathfrak{g}_i$ are the projections, the sum identities give surjectivity and the intersection identities give injectivity. The restrictions $p_i|_\mathfrak{h} : \mathfrak{h} \to \mathfrak{g}_i$ are therefore isomorphisms. Since $\mathfrak{g}_1$ is simple, $\mathfrak{h}$ is simple, and so is $\mathfrak{g}_2$. After an automorphism the pair is $(\mathfrak{l} \oplus \mathfrak{l}, \mathrm{diag}\, \mathfrak{l})$. Its quotient is the adjoint $\mathfrak{l}$-module, whose real endomorphism algebra is $\mathbb{R}$, again contradicting (1). This contradiction proves that $\mathfrak{g}$ is simple.

Semisimplicity of $\mathfrak{h}$. For $0 \neq z \in \mathfrak{z}(\mathfrak{h})$, Schur's lemma gives $T_z = \lambda I$ for some $\lambda \in \mathbb{R}$, and faithfulness gives $\lambda \neq 0$. For $0 \neq X \in \mathrm{Rad}(c_V)$, faithfulness gives $T_X \neq 0$, while equation (2) would give

$$0 = c_V(X, z, X) = \lambda \operatorname{Tr}_V(T_X^2),$$

which is impossible because the nonzero Hermitian operator T_X satisfies $\operatorname{Tr}_V(T_X^2) > 0$. Consequently $\mathfrak{z}(\mathfrak{h}) = 0$; compactness then makes $\mathfrak{h}$ semisimple. □

Remark (Strength of the mixed condition). The mixed arguments are essential. If only $c_V(\mathfrak{k}, \mathfrak{k}, \mathfrak{k}) = 0$ were imposed, the pairs $\mathfrak{e}_6 \supset \mathfrak{so}(10) \oplus \mathfrak{u}(1)$ and $\mathfrak{e}_7 \supset \mathfrak{e}_6 \oplus \mathfrak{u}(1)$ would satisfy it for $\mathfrak{k} = \mathfrak{so}(10)$ and $\mathfrak{k} = \mathfrak{e}_6$, respectively; neither satisfies (3).

3 Exhaustive census of the homogeneous domain

Lemma 2.2 narrows no hypothesis and loses no pair. It puts every object in the original classification domain into a forced structural normal form: $\mathfrak{g}$ is compact and simple, $\mathfrak{h}$ is maximal semisimple, and the complement is irreducible of complex type. The proof below exhausts the full-rank and rank-deficient branches of that original domain. The first becomes a finite root-grading calculation. The second is eliminated by exact index arithmetic in exceptional ambient algebras and by a highest-weight argument in the classical ones.

We use $A_n = \mathfrak{su}(n+1)$. A subalgebra is called regular when it is normalized by a Cartan subalgebra of $\mathfrak{g}$.

3.1 The full-rank branch

A regular maximal subalgebra has full rank. Indeed, if $\operatorname{rank} \mathfrak{h} < \operatorname{rank} \mathfrak{g}$, regularity supplies a nonzero toral subalgebra $\mathfrak{t}_0 \subset C_{\mathfrak{g}}(\mathfrak{h})$, and

$$\mathfrak{h} \subsetneq \mathfrak{h} + \mathfrak{t}_0 \subsetneq \mathfrak{g},$$

contrary to maximality. Conversely, any equal-rank subalgebra contains a maximal torus of $\mathfrak{g}$ and is regular.

Lemma 3.1 (Saturation of a closed root subsystem)**.** *If $\Psi \subseteq \Phi$ is a closed root subsystem, then*

$$\Phi \cap \mathbb{Z}\Psi = \Psi.$$

In particular, if $\Psi \subsetneq \Phi$, then $\mathbb{Z}\Psi \neq \mathbb{Z}\Phi$.

Proof. Let $\alpha \in \Phi \cap \mathbb{Z}\Psi$. Write $\alpha = \beta_1 + \cdots + \beta_m$, with each $\beta_j \in \Psi$, and choose such an expression with m minimal; negative coefficients may be absorbed by replacing roots of Ψ with their negatives. The claim is immediate for $m = 1$. If $m > 1$, then

$$(\alpha, \alpha) = \sum_{j=1}^{m} (\alpha, \beta_j) > 0,$$

so $(\alpha, \beta_j) > 0$ for some j. If $\alpha = \beta_j$, then $\alpha \in \Psi$. Otherwise the root-string property gives $\alpha - \beta_j \in \Phi$. This root lies in $\mathbb{Z}\Psi$ and is a sum of $m - 1$ roots of Ψ, hence belongs to Ψ by induction. Closedness of Ψ then gives $\alpha = (\alpha - \beta_j) + \beta_j \in \Psi$. Thus $\Phi \cap \mathbb{Z}\Psi \subseteq \Psi$; the reverse inclusion is immediate. If $\mathbb{Z}\Psi = \mathbb{Z}\Phi$, the displayed equality gives $\Phi = \Psi$, proving the final assertion. □

Lemma 3.2 (Complex type and cyclic root gradings)**.** *Suppose* $\operatorname{rank}\mathfrak{h} = \operatorname{rank}\mathfrak{g}$. *Then the root spaces of* $\mathfrak{g}_{\mathbb{C}}$ *relative to a common Cartan subalgebra admit one of the following two gradings:*

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{h}_{\mathbb{C}} \oplus \mathfrak{g}_1 \quad (\mathbb{Z}_2), \qquad \textit{or} \qquad \mathfrak{g}_{\mathbb{C}} = \mathfrak{h}_{\mathbb{C}} \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \quad (\mathbb{Z}_3),$$

where in the second case $\mathfrak{g}_1 \cong V$, $\mathfrak{g}_2 \cong V^*$, *and* $[\mathfrak{g}_i, \mathfrak{g}_j] \subseteq \mathfrak{g}_{i+j}$. *The involutive case has a self-conjugate quotient and therefore cannot satisfy condition* (1).

Proof. Let Φ be the root system of $\mathfrak{g}_{\mathbb{C}}$, let Ψ be the root subsystem of $\mathfrak{h}_{\mathbb{C}}$, and put $Q = \mathbb{Z}\Phi$, $L = \mathbb{Z}\Psi$. Root operators of $\mathfrak{h}$ change weights by elements of L. Since V is irreducible, all roots whose root spaces lie in V have one class $c \in Q/L$; those in V^* have class $-c$. Every root therefore has class $0, c$, or $-c$, and these classes generate Q/L. Since Ψ is proper and closed, Lemma 3.1 gives $Q/L \neq 0$. Hence $c \neq 0$.

If $2c = 0$, the quotient is the odd part of an inner involution. Compact conjugation preserves that odd root set because $-c = c$. For a classical ambient algebra, an inner involution either has a unitary block centralizer with nonzero center, or has an orthogonal or symplectic block centralizer whose odd module is a self-dual tensor product. For an exceptional ambient algebra, the exact involutive census in Appendix A shows that whenever the fixed algebra is semisimple, the odd weight graph is connected. In every case compatible with semisimplicity the irreducible odd module is self-conjugate, not of complex type.

Assume now that $2c \neq 0$. If $[V, V] = 0$, compact conjugation also gives $[V^*, V^*] = 0$, while the root classes give $[V, V^*] \subseteq \mathfrak{h}_{\mathbb{C}}$. The map acting as $0, +1, -1$ on $\mathfrak{h}_{\mathbb{C}}, V, V^*$, respectively, is then a nonzero derivation of the simple algebra $\mathfrak{g}_{\mathbb{C}}$. It is inner and centralizes $\mathfrak{h}_{\mathbb{C}}$. But $\mathfrak{h}$ is semisimple of full rank, so its roots span the dual of the common Cartan and its centralizer in $\mathfrak{g}_{\mathbb{C}}$ is zero, a contradiction. Hence $[V, V] \neq 0$. A nonzero bracket has root class $2c$, which must be one of $0, c, -c$. The first possibility is excluded by $2c \neq 0$, and the second would give $c = 0$. Thus $2c = -c$, so $3c = 0$. The quotient map $Q \to Q/L \cong \mathbb{Z}_3$ defines a toral order-three automorphism with fixed algebra $\mathfrak{h}_{\mathbb{C}}$ and the asserted grading. □

Proposition 3.3 (Direct toral grading census)**.** *Let* (G, H) *lie in the compact homogeneous domain defined above: the standing pair hypotheses together with Conditions* (1) *and* (3). *By Lemma 2.2, without any additional hypothesis,* $\mathfrak{g}$ *is simple,* $\mathfrak{h}$ *is maximal semisimple, and the complement is irreducible of complex type. In the full-rank case the pair is isomorphic to exactly one of the following six types:*

$$\mathfrak{g}_2 \supset A_2, \quad \mathfrak{f}_4 \supset A_2 + A_2, \quad \mathfrak{e}_6 \supset A_2^3, \quad \mathfrak{e}_7 \supset A_5 + A_2, \quad \mathfrak{e}_8 \supset A_8, \quad \mathfrak{e}_8 \supset E_6 + A_2.$$

Their complex halves have dimensions

$$3, \quad 18, \quad 27, \quad 45, \quad 84, \quad 81,$$

respectively, and are irreducible and inequivalent to their duals.

Proof. Condition (3) has entered this candidate list only through Lemma 2.2, where it forces $\mathfrak{h}$ to be semisimple. No cubic coefficient has yet been used to eliminate a row; that selection follows after the rank-deficient branch is closed.

For the classical groups the calculation is immediate in the defining module. A nonidentity inner automorphism of order three of $\mathfrak{su}(n)$ has a representative with eigenvalues among $c, c\zeta, c\zeta^2$, where $\zeta^3 = 1$, $\zeta \neq 1$; the representative need not have order three. If $s \geq 2$ eigenspaces are nonempty, its centralizer $S(\prod_{j=1}^{s} U(n_j))$ has center dimension $s - 1 > 0$. In $SO(n)$ and $Sp(n)$, an order-three

inner automorphism admits an order-three representative, since their centers have order coprime to three. A nonidentity such representative has a unitary factor in its centralizer, again giving a center. Thus no proper classical fixed algebra is semisimple. For involutions, the semisimple classical centralizers have self-dual tensor-product quotients.

For the exceptional groups the remaining calculation is finite. Appendix A enumerates every nonzero homomorphism $Q \to \mathbb{Z}_m$, $m = 2, 3$, modulo the Weyl action, directly from the Cartan matrices. It retains the gradings whose grade-zero roots span the full rank and tests connectedness of the grade-one weight graph. For $m = 2$ every semisimple row has a connected self-conjugate odd module. For $m = 3$ the six rows displayed above are the complete list. The Killing form pairs grades one and two nondegenerately, so the two modules are dual; the connected grade-one graphs prove irreducibility. □

Lemma 3.4 (Degree-three invariant theory)**.** *For a simple complex Lie algebra* $\mathfrak{s}$*,*

$$\left(\mathrm{Sym}^3 \mathfrak{s}^*\right)^{\mathfrak{s}} \neq 0 \quad \Longleftrightarrow \quad \mathfrak{s} \text{ has type } A_n \text{ with } n \geq 2.$$

Thus only type A factors can carry a nonzero perturbative cubic coefficient; a particular representation of such a factor may still have coefficient zero.

Proof. By Chevalley's theorem, invariant polynomials on $\mathfrak{s}$ have basic degrees equal to the exponents plus one. Degree three occurs precisely in type A_n, $n \geq 2$. □

3.2 The rank-deficient branch

No classification of maximal subalgebras is needed in this branch. The exceptional ambient algebras are excluded by a finite arithmetic sieve; the classical ambient algebras admit a uniform highest-weight argument.

Proposition 3.5 (Exceptional rank-deficient no-go)**.** *Let* $\mathfrak{g}$ *be exceptional simple and let* $\mathfrak{h} \subset \mathfrak{g}$ *be semisimple with* $\operatorname{rank}\mathfrak{h} < \operatorname{rank}\mathfrak{g}$*. If* $(\mathfrak{g}/\mathfrak{h})_{\mathbb{C}} = V \oplus V^*$ *with* V *irreducible of complex type, then* $\mathrm{Rad}(c_V) = 0$*.*

Proof. Write $\mathfrak{h} = \bigoplus_i \mathfrak{h}_i$ and $V = \boxtimes_i V_i$. Let $d_i = \dim V_i$. Normalize the quadratic index by taking long roots to have squared length two and requiring $T(\mathrm{ad}_{\mathfrak{h}_i}) = h_i^\vee$; in particular $T_{A_{n-1}}(\mathbf{n}) = \frac{1}{2}$. If $I_i \in \mathbb{Z}_{>0}$ is the Dynkin index of $\mathfrak{h}_i \hookrightarrow \mathfrak{g}$, restriction of the adjoint representation gives the necessary equations

$$\dim \mathfrak{g} = \sum_i \dim \mathfrak{h}_i + 2\prod_i d_i, \qquad h_{\mathfrak{g}}^\vee I_i = h_i^\vee + 2T_i \prod_{j\neq i} d_j. \tag{6}$$

The normalization is visible in the surviving full-rank row. For $E_8 \supset E_6 + A_2$, both embeddings have index one and

$$30 = 12 + 2 \cdot 3 \cdot 3, \qquad 30 = 3 + 2 \cdot \frac{1}{2} \cdot 27. \tag{6a}$$

The radical is nonzero only if at least one simple factor has vanishing cubic coefficient on V_i; Lemma 3.4 makes this condition automatic for every non-A factor.

The dimension equation bounds the representation search before any index is computed. Since $\mathfrak{h} \neq 0$ is semisimple, rank deficient, and $\dim \mathfrak{g} - \dim \mathfrak{h} = 2 \dim V$, the smallest parity-compatible possibility for $\dim \mathfrak{h}$ gives

$\mathfrak{g}$	G_2	F_4	E_6	E_7	E_8
smallest compatible $\dim \mathfrak{h}$	none	6	6	3	6
$\max \dim V$	none	23	36	65	121.

(6b)

For G_2, the only smaller-rank semisimple type is A_1, and $14-3$ is odd. For the even-dimensional ambient algebras the smallest parity-compatible semisimple dimension is six, while E_7 admits A_1 of dimension three.

There are therefore finitely many factor types; for each one, Weyl's dimension formula is strictly increasing in every Dynkin label, so only finitely many dominant weights occur. Appendix A enumerates all of them inside the bounds (6b), quotients permutations of identical factors, and imposes complex type, a nonzero radical factor, the dimension equation, and the integrality of every equation in (6). Its audit records every candidate reaching the index stage. No candidate remains for G_2, F_4, E_6, E_7, or E_8. □

Lemma 3.6 (Defining modules and canonical constituents)**.** *Let $\mathfrak{g}_{\mathbb{C}}$ be $\mathfrak{sl}(W)$, $\mathfrak{so}(W)$, or $\mathfrak{sp}(W)$, and suppose $\mathfrak{h}_{\mathbb{C}} \subset \mathfrak{g}_{\mathbb{C}}$ is semisimple and maximal with quotient $V \oplus V^*$, where $V \ncong V^*$ is irreducible. Then W is irreducible as an $\mathfrak{h}_{\mathbb{C}}$-module and $\mathfrak{h}_{\mathbb{C}}$ is simple.*

If λ is the highest weight of W, let λ^* be the highest weight of W^*, and write $i \mapsto i^*$ for the Dynkin-node involution induced by duality. The following constituents occur with multiplicity one:

$$V_{\lambda+\lambda^*} \subset W \otimes W^*, \qquad V_{2\lambda} \subset \operatorname{Sym}^2 W,$$

and, when W is orthogonal,

$$V_{2\lambda-\alpha_i} \subset \Lambda^2 W \quad \text{for every } i \text{ with } \langle \lambda, \alpha_i^\vee \rangle > 0. \tag{6c}$$

If the last integer is at least three, then $V_{2\lambda-3\alpha_i}$ also occurs in $\Lambda^2 W$.

Proof. Suppose first that the defining module is reducible. In the compact real form, complete reducibility gives either a proper invariant orthogonal decomposition or an irreducible real module whose complexification splits as an isotropic dual pair. In the first case let B be the compact block stabilizer. If $\mathfrak{h} \subsetneq B$, maximality is contradicted. If $\mathfrak{h} = B$, then in the unitary case B has nonzero center, contrary to semisimplicity; in the orthogonal and symplectic cases the off-diagonal quotient is a sum of self-dual tensor products of defining blocks, contrary to irreducible complex type. Under the isotropic-dual-pair alternative, the splitting defines an $\mathfrak{h}$-invariant orthogonal complex structure, and $\mathfrak{h}$ lies in its proper compact unitary centralizer, contradicting maximality. Every possibility is excluded, so the defining module is irreducible, and $W = \boxtimes_i W_i$ for the simple ideals of $\mathfrak{h}$. In $\mathfrak{sl}(W)$, two distinct factors produce the self-dual constituent $\operatorname{ad}\mathfrak{h}_i \boxtimes \operatorname{ad}\mathfrak{h}_j$ outside $\mathfrak{h}$. In the orthogonal or symplectic cases every W_i is self-dual. Writing $\operatorname{End}(W_i) = K_i \oplus P_i$ for the skew- and self-adjoint parts, the ambient Lie algebra is the odd-parity part of $\bigotimes_i (K_i \oplus P_i)$. Three factors again give the self-dual constituent $\operatorname{ad}\mathfrak{h}_1 \boxtimes \operatorname{ad}\mathfrak{h}_2 \boxtimes \operatorname{ad}\mathfrak{h}_3$. With two factors, the nonzero blocks $\operatorname{ad}\mathfrak{h}_1 \boxtimes P_2^0$ and $P_1^0 \boxtimes \operatorname{ad}\mathfrak{h}_2$ lie in disjoint tensor summands outside $\mathfrak{h}$ and are self-dual by their nondegenerate trace pairings. But $V \oplus V^*$ has no nonzero proper self-dual submodule, so it cannot contain both blocks. If $P_1^0 = 0$, then W_1 is two-dimensional symplectic. When $P_2^0 \neq 0$, maximality forces $\mathfrak{h}_2 = K_2$, and the quotient is $\operatorname{ad}\mathfrak{sl}_2 \boxtimes P_2^0$, an irreducible self-dual module; if both traceless parts vanish, the quotient is zero. Hence $\mathfrak{h}$ is simple.

The first two displayed inclusions are the Cartan products. For the third, if v_λ is a highest-weight vector and f_i a simple lowering operator, then $v_\lambda \wedge f_i v_\lambda$ is a highest-weight vector of weight $2\lambda - \alpha_i$. If $m = \langle \lambda, \alpha_i^\vee \rangle \geq 3$, the vector

$$m\, v_\lambda \wedge f_i^3 v_\lambda - 3(m-2) f_i v_\lambda \wedge f_i^2 v_\lambda$$

is nonzero and highest of weight $2\lambda - 3\alpha_i$. In the remaining dual-node cases, this weight and its dual are distinct from the adjoint and from the pair $2\lambda - \alpha_i, 2\lambda - \alpha_{i^*}$; they therefore supply an additional quotient pair. □

Proposition 3.7 (Classical rank-deficient no-go)**.** *Let (G, H) satisfy the standing pair hypotheses, let $\mathfrak{g}$ be classical simple, and let $\mathfrak{h} \subset \mathfrak{g}$ be rank deficient. Under condition* (1)*, the radical of the complex half is zero.*

Proof. Assume for contradiction that $\mathrm{Rad}(c_V) \neq 0$. Then the standing pair hypotheses, condition (1), and Lemma 2.2 imply that $\mathfrak{h}$ is semisimple and maximal, so Lemma 3.6 applies; use its notation. If $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}(W)$, the self-dual Cartan constituent $V_{\lambda+\lambda^*}$ must be the adjoint module of $\mathfrak{h}$; otherwise it would be a self-dual summand of the quotient. Thus $\lambda + \lambda^* = \theta$, where θ is the highest root of $\mathfrak{h}_{\mathbb{C}}$. The three Cartan-product equations used in this proof are labelwise, and their nonzero dominant solutions are

$$\begin{array}{ll} \lambda + \lambda^* = \theta & A_r : \ \omega_1, \omega_r; \quad C_r : \ \omega_1 \text{ (equivalently } B_2 : \ \omega_2), \\ 2\lambda = \theta & C_r : \ \omega_1 \text{ (including } A_1, \text{ and equivalently } B_2 : \ \omega_2), \\ 2\lambda - \alpha_i = \theta, \ i = i^* & A_1 : \ 2\omega_1; \ A_3 : \ \omega_2; \ B_r : \ \omega_1; B_3 : \ \omega_3; \ C_2 : \ \omega_2; \\ & D_r : \ \omega_1; \ D_4 : \ \omega_3, \omega_4; \ G_2 : \ \omega_1. \end{array} \tag{6d}$$

The first row gives the standard A_r representation, for which the ambient algebra equals $\mathfrak{h}$, and the defining symplectic representation, whose quotient in $\mathfrak{sl}(W)$ is self-dual. The second row gives only the standard symplectic algebra and low-dimensional isomorphisms, so $\mathfrak{sp}(W)$ has no proper complex-type quotient.

Suppose $\mathfrak{g}_{\mathbb{C}} = \mathfrak{so}(W)$. For every support node fixed by duality, the constituent in (6c) is self-dual and must equal the adjoint, so $2\lambda - \alpha_i = \theta$. The third row of (6d) consists of defining representations and low-rank isomorphisms, together with $G_2 \subset SO(7)$ and the spin embedding $B_3 \subset SO(8)$; their quotients are zero or self-dual. Consequently the support of λ, if any case remains, is one pair of nodes exchanged by duality, with a common coefficient a. This occurs only in types A_r, odd D_r, and E_6.

For A_{N-1}, choose the pair $i, N-i$ with $1 \leq i < N/2$. The constituent $\mu_i = 2\lambda - \alpha_i$ has cubic Harish–Chandra eigenvalue

$$C_3(\mu_i) = -6a(2a + N - 2i) \neq 0. \tag{6e}$$

It is therefore one of the two quotient halves with nonzero cubic index, so the radical is zero.

For odd D_r and for E_6, a coefficient $a \geq 3$ gives the additional constituent $V_{2\lambda - 3\alpha_i}$, contradicting the assumption that the quotient has only one dual pair. It remains to inspect $a = 1, 2$. Put

$$D_{r,a} = \dim V_{a(\omega_{r-1}+\omega_r)}, \qquad M_{r,a} = \dim V_{2a(\omega_{r-1}+\omega_r)-\alpha_{r-1}}.$$

Weyl's formula gives, for odd $r \geq 5$,

$$\frac{4M_{r,1}}{D_{r,1}^2} = \frac{36(r-1)(2r+1)}{r(r+2)(r+3)(r+4)},$$

$$\frac{4M_{r,2}}{D_{r,2}^2} = \frac{5{,}376{,}000(r-1)(r+2)(2r+1)(2r+3)^2(2r+5)}{r(r+1)(r+3)^2(r+4)^3(r+5)^3(r+6)^2(r+7)(r+8)}.$$

Set $R_{r,a} = 4M_{r,a}/D_{r,a}^2$. The two sequences decrease for $r \geq 5$. Indeed, after clearing the positive denominators in $R_{r+1,a}/R_{r,a} < 1$ and writing $n = r - 5$, the two differences are

$$4n^3 + 57n^2 + 259n + 365$$

and

$$\begin{aligned} &32n^7 + 1{,}624n^6 + 34{,}692n^5 + 404{,}090n^4 \\ &\quad + 2{,}769{,}860n^3 + 11{,}162{,}622n^2 + 24{,}455{,}216n + 22{,}419{,}744, \end{aligned}$$

Table 1: The direct full-rank census and its cubic coefficients. The radical is the sum of the factors with zero coefficient.

pair $\mathfrak{g} \supset \mathfrak{h}$	complex summand V	$\mathbf{A}(V)$	$\mathrm{Rad}(c_V)$
$\mathfrak{g}_2 \supset \mathfrak{su}(3)$	$\mathbf{3}$	1	0
$\mathfrak{f}_4 \supset \mathfrak{su}(3) \oplus \mathfrak{su}(3)$	$\mathbf{6} \boxtimes \overline{\mathbf{3}}$	$(21, -6)$	0
$\mathfrak{e}_6 \supset \mathfrak{su}(3)^{\oplus 3}$	$\mathbf{3} \boxtimes \mathbf{3} \boxtimes \mathbf{3}$	$(9, 9, 9)$	0
$\mathfrak{e}_7 \supset \mathfrak{su}(6) \oplus \mathfrak{su}(3)$	$\mathbf{15} \boxtimes \overline{\mathbf{3}}$	$(6, -15)$	0
$\mathfrak{e}_8 \supset \mathfrak{su}(9)$	$\Lambda^3\mathbf{9}$	9	0
$\mathfrak{e}_8 \supset \mathfrak{e}_6 \oplus \mathfrak{su}(3)_M$	$\mathbf{27} \boxtimes \mathbf{3}$	$(0, 27)$	$\mathfrak{e}_6$

respectively. Hence

$$R_{r,1} \le R_{5,1} = \frac{22}{35} < \frac{2}{3}, \qquad R_{r,2} \le R_{5,2} = \frac{1274}{8019} < \frac{1}{6}.$$

Moreover $D_{r,1} = \binom{2r}{r-1} \ge r(2r-1)$, and monotonicity in the highest weight gives $D_{r,2} > D_{r,1}$. Writing $H = r(2r-1)$, we obtain

$$\binom{D_{r,1}}{2} - H - 2M_{r,1} \ge \frac{H(H-9)}{6} > 0,$$
$$\binom{D_{r,2}}{2} - H - 2M_{r,2} \ge \frac{H(5H-18)}{12} > 0.$$

Thus $\Lambda^2 W$ contains more than the adjoint and one dual pair. For E_6, the two possible supports are $\{\omega_1, \omega_6\}$ and $\{\omega_3, \omega_5\}$; the four remaining values $a = 1, 2$ give the positive dimension gaps

$$52{,}897, \quad 3{,}008{,}865{,}288, \quad 1{,}949{,}250{,}433, \quad 24{,}307{,}954{,}884{,}506{,}865.$$

The exact Dynkin-label and dimension calculations are reproduced in Appendix A. No rank-deficient classical pair can have a nonzero radical. □

3.3 Cubic indices and the survivor

For a simple factor of type A_{N-1} with $N \ge 3$, let $A(R)$ be the cubic index, normalized by $A(\mathbf{N}) = 1$; for A_1 and every other simple type put $A(R) = 0$. Duality gives $A(R^*) = -A(R)$, and for $V = \boxtimes_i V_i$ the coefficient on the i-th factor is

$$A_i(V) = A(V_i) \prod_{j \ne i} \dim V_j. \tag{6f}$$

Write $\mathbf{A}(V) := (A_i(V))_i$. Mixed cubics between distinct simple ideals vanish because they factor through the trace of a generator. Hence $\mathrm{Rad}(c_V)$ is the sum of the simple ideals whose coefficients in (6f) vanish.

The six full-rank rows of Proposition 3.3 give:

The entries follow, for $N \ge 3$ and $1 \le k \le N-1$, from

$$A(\Lambda^k \mathbf{N}) = \frac{(N-2k)(N-3)!}{(k-1)!(N-k-1)!}, \qquad A(\mathrm{Sym}^2 \mathbf{N}) = N + 4,$$

and the tensor-product rule (6f); the trivial exterior powers at $k = 0, N$ have index zero. The Lie algebra E_6 has no invariant symmetric cubic, whereas A_2 does. The order-three grading produces the pair $E_6 + A_2$; degree-three invariant theory distinguishes the anomalous factor from the radical. Thus the last row has

$$\mathbf{A}(V) = (0, 27), \qquad \mathrm{Rad}(c_V) = \mathfrak{e}_6.$$

The cubic on the **27** is a trilinear form on the representation; the vanishing used here concerns invariant trilinear forms on the Lie algebra $\mathfrak{e}_6$.

Proof of Theorem 2.1. Lemma 2.2 places every pair in the homogeneous domain into the same structural normal form. Lemma 3.2 and Proposition 3.3 give the complete full-rank branch; Table 1 leaves only its final row. Propositions 3.5 and 3.7 exclude every rank-deficient pair with nonzero radical. The final row satisfies all hypotheses, proving the converse. □

Remark (Classification closure). The proof uses no classification of maximal subalgebras or homogeneous spaces. The full-rank branch is a finite toral-grading census from Cartan matrices; the exceptional rank-deficient branch is a finite Weyl-dimension/index sieve; and the classical rank-deficient branch is the uniform highest-weight argument above. The manuscript supplies the reductions and completeness criteria; the distributed programs discharge the resulting finite leaves.

3.4 The centralizer and multiplicity three

The surviving zero coefficient has an immediate multiplicity interpretation. For the selected pair,

$$\mathrm{Rad}(c_V) = \mathfrak{e}_6, \qquad C_{\mathfrak{h}}(\mathrm{Rad}(c_V)) = \mathfrak{su}(3)_M.$$

Here $C_{\mathfrak{h}}$ denotes the centralizer in $\mathfrak{h}$. The radical is the largest ideal on which all cubic coefficients, including mixed ones, vanish. Its centralizer acts on the second tensor factor, and its defining **3** gives three copies of the **27** after restriction to E_6. The subscript M records this multiplicity. The construction uses $SU(3)_M$ as a commuting multiplicity factor, not as a gauged symmetry; Section 5 computes its cubic anomaly.

The ambient algebra is E_8. Lorentz symmetry remains external, and matter is the complex summand V, not the adjoint representation. This differs from adjoint-based E_8 unification, in which gauge and Lorentz symmetries are embedded in a real or complex form of E_8 [28]. The adjoint branching displays V and V^* as the two halves of the homogeneous complement:

$$\mathbf{248} = (\mathbf{78}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{8}) \oplus (\mathbf{27}, \mathbf{3}) \oplus (\overline{\mathbf{27}}, \overline{\mathbf{3}}). \tag{7}$$

Restriction to the first factor gives $V|_{E_6^{\mathrm{sc}}} = 3\,\mathbf{27}$ and $V^*|_{E_6^{\mathrm{sc}}} = 3\,\overline{\mathbf{27}}$. Either orientation therefore has multiplicity three.

3.5 The global exceptional subgroup

The same branching also determines the global image inside E_8. The Lie-algebra inclusion integrates to a homomorphism $\varphi : E_6^{\mathrm{sc}} \times SU(3)_M \to E_8$ with finite central kernel. Choose generators $z_E \in Z(E_6^{\mathrm{sc}})$ and $z_3 \in Z(SU(3)_M)$ that act by the same primitive cube root ζ_3 on **27** and **3**. The adjoint decomposition (7) shows that $\langle (z_E, z_3^{-1}) \rangle$ acts trivially on every summand. Conversely, a central kernel element lies in the product of the two order-three centers, and its action on $\mathbf{27} \boxtimes \mathbf{3}$ forces it into this diagonal subgroup. Faithfulness of the adjoint representation of E_8 leaves the kernel $\langle (z_E, z_3^{-1}) \rangle$. The global subgroup is

$$\frac{E_6^{\mathrm{sc}} \times SU(3)_M}{\langle (z_E, z_3^{-1}) \rangle} \subset E_8.$$

The image of $(z_E, 1)$ acts on $\mathfrak{h}_{\mathbb{C}}, V, V^*$ with eigenvalues $1, \zeta_3, \zeta_3^{-1}$, respectively, and realizes (7) as a $\mathbb{Z}/3$-grading. The branching is classical [14, 16]; the classification above shows that it is the only row satisfying (1) and (3).

4 Extremal intersections in E_6

The homogeneous classification leaves a single embedded E_6^{sc}, up to E_8-conjugacy. Theorem 2.1 fixes the pair $E_6 + A_2 \subset E_8$ up to automorphisms of $\mathfrak{e}_8$, and all automorphisms of compact $\mathfrak{e}_8$ are inner.

We now work inside this E_6. Closed root subsystems determine connected regular subgroups, and their intersections determine the semisimple roots of common connected subgroups. Choosing one maximal proper subsystem from each Weyl orbit gives a finite configuration space. We study two Weyl-invariant questions on it: simultaneous pairwise maximality and maximal common intersection.

The calculation rests on three pairwise facts. A D_5–$(A_5 + A_1)$ intersection has two possible types; the D_5–A_2^3 intersection is rigid; and an $(A_5 + A_1)$–A_2^3 intersection is large precisely when the distinguished root pair lies in the A_2^3 subsystem. The two extremal theorems combine these facts in different ways.

4.1 The finite configuration space

The three maximal subsystem types admit concrete parameterizations that make the finite problem explicit. Fix a maximal torus $T_0 \subset E_6^{\mathrm{sc}}$ with Lie algebra $\mathfrak{t}_0$, let $\Phi = \Phi(E_6)$, and write $\mathcal{W} = W(E_6)$. A subsystem $\Psi \subseteq \Phi$ is *closed* when

$$\alpha, \beta \in \Psi, \quad \alpha + \beta \in \Phi \quad \Longrightarrow \quad \alpha + \beta \in \Psi.$$

For such Ψ, let $\widehat{\Psi}$ be the connected regular subgroup containing T_0 with

$$\mathrm{Lie}(\widehat{\Psi})_{\mathbb{C}} = (\mathfrak{t}_0)_{\mathbb{C}} \oplus \bigoplus_{\beta \in \Psi} (\mathfrak{e}_6)_\beta.$$

Proposition 4.1 (Maximal closed subsystems of E_6)**.** *The maximal proper closed subsystems of Φ form exactly three $\mathcal{W}$-orbits*

$$\mathcal{O}_D \ (D_5), \qquad \mathcal{O}_A \ (A_5 + A_1), \qquad \mathcal{O}_\Theta \ (A_2^3).$$

Let ω_1 be a minuscule fundamental weight, whose Weyl orbit has 27 elements, and put $\Lambda_{27} = \mathcal{W}\omega_1$, and let $\mathcal{R} = \Phi/\{\pm 1\}$. Here $\beta^\vee$ denotes the coroot of β. Then

$$\begin{aligned} D_\lambda &= \{\beta \in \Phi : \langle \lambda, \beta^\vee \rangle = 0\}, \quad \lambda \in \Lambda_{27}, \\ A_{\bar{\alpha}} &= \{\pm\alpha\} \cup (\Phi \cap \alpha^\perp), \qquad \bar{\alpha} = \{\pm\alpha\} \in \mathcal{R}, \end{aligned}$$

give $\mathcal{W}$-equivariant bijections $\Lambda_{27} \simeq \mathcal{O}_D$ and $\mathcal{R} \simeq \mathcal{O}_A$.

Proof. The exact closed-subsystem traversal of Appendix B gives $D_5, A_5 + A_1, A_2^3$ as the maximal proper types in E_6. Roots orthogonal to a minuscule weight form D_5; a root pair $\{\pm\alpha\}$ determines the A_1 that it spans together with the orthogonal A_5. If $D_\lambda = D_\mu$, their one-dimensional orthogonal complements give $\mu = \pm\lambda$; the negative weights lie in the dual minuscule orbit, so $\mu = \lambda$. Likewise, $A_{\bar{\alpha}}$ recovers $\{\pm\alpha\}$ as its A_1-component. The orbit classification and these injectivity statements give both bijections. □

Remark (The mark-three subsystem). The orbit $\mathcal{O}_\Theta$ is the Weyl orbit of the fixed-root subsystem of the mark-three toral grading of E_6 in Proposition 3.3. The two inverse nontrivial characters interchange the grade-one and grade-two modules, identifying them with the mutually dual 27-dimensional A_2^3-modules displayed in the adjoint branching below. They are modules for the grade-zero algebra A_2^3, not the fundamental **27** and $\overline{\mathbf{27}}$ of E_6.

4.2 The two extremal orbits

The first criterion maximizes every pairwise intersection; the second maximizes the threefold intersection itself. Put

$$\mathcal{X} = \mathcal{O}_D \times \mathcal{O}_A \times \mathcal{O}_\Theta.$$

For subsystems P, Q drawn from distinct orbit factors, call $P \cap Q$ maximal when $|P \cap Q|$ attains its maximum over the full product of those two orbits. Because

$$\dim_{\mathbb{R}}((\widehat{P} \cap \widehat{Q})^\circ) = 6 + |P \cap Q|,$$

maximizing $|P \cap Q|$ is equivalent to maximizing the dimension of the identity component. Let $\mathcal{X}_{\max}$ be the set of triples for which all three pairwise intersections are largest.

Theorem 4.2 (Triples with maximal pairwise intersections)**.** *The pairwise intersections are*

pair	*root-system type*		$\lvert P \cap Q \rvert$
$D_\lambda \cap A_{\bar\alpha}$	A_4,	$\langle \lambda, \alpha^\vee \rangle \neq 0$,	20
	$A_3 + A_1 + A_1$,	$\langle \lambda, \alpha^\vee \rangle = 0$	16
$D_\lambda \cap \Theta$	$A_2 + A_1 + A_1$		10
$A_{\bar\alpha} \cap \Theta$	$A_2 + A_2 + A_1$,	$\{\pm\alpha\} \subset \Theta$,	14
	$A_1 + A_1 + A_1$,	$\{\pm\alpha\} \not\subset \Theta$	6

and therefore

$$\mathcal{X}_{\max} = \{(D_\lambda, A_{\bar\alpha}, \Theta) : \langle \lambda, \alpha^\vee \rangle \neq 0,\ \{\pm\alpha\} \subset \Theta\}.$$

For every $s = (D_\lambda, A_{\bar\alpha}, \Theta) \in \mathcal{X}_{\max}$,

$$D_\lambda \cap A_{\bar\alpha} \cong A_4, \qquad R_s := D_\lambda \cap A_{\bar\alpha} \cap \Theta \cong A_2 + A_1.$$

Its stabilizer and orbit size are

$$\mathrm{Stab}_{\mathcal{W}}(s) = W(R_s), \qquad \mathcal{X}_{\max} \simeq \mathcal{W}/W(R_s), \qquad |\mathcal{X}_{\max}| = 4320.$$

Thus $\mathcal{X}_{\max}$ *is a single* $\mathcal{W}$*-orbit.*

Proof. Pairwise types. The required branchings [14, 15] are

$$\begin{aligned}
\mathbf{27}|_{A_5+A_1} &= (\mathbf{15},\mathbf{1}) \oplus (\overline{\mathbf{6}},\mathbf{2}),\\
\mathbf{27}|_{A_2^3} &= (\mathbf{3},\overline{\mathbf{3}},\mathbf{1}) \oplus (\mathbf{1},\mathbf{3},\overline{\mathbf{3}}) \oplus (\overline{\mathbf{3}},\mathbf{1},\mathbf{3}),\\
\mathfrak{e}_6|_{A_2^3} &= (\mathbf{8},\mathbf{1},\mathbf{1}) \oplus (\mathbf{1},\mathbf{8},\mathbf{1}) \oplus (\mathbf{1},\mathbf{1},\mathbf{8})\\
&\quad \oplus (\mathbf{3},\mathbf{3},\mathbf{3}) \oplus (\overline{\mathbf{3}},\overline{\mathbf{3}},\overline{\mathbf{3}}).
\end{aligned}$$

In the first branching, a weight in $(\mathbf{15},\mathbf{1})$ has orthogonal roots $A_3 + A_1$ inside A_5, together with the external A_1; this gives $A_3 + A_1 + A_1$. A weight in $(\overline{\mathbf{6}},\mathbf{2})$ instead has orthogonal roots A_4. These

are respectively the cases $\langle \lambda, \alpha^\vee \rangle = 0$ and $\langle \lambda, \alpha^\vee \rangle \neq 0$. In the second branching, every weight is nontrivial on two A_2-factors and trivial on the third. A weight of $\mathbf{3}$ or $\overline{\mathbf{3}}$ is orthogonal to one root pair in A_2, so $D_\lambda \cap \Theta \cong A_2 + A_1 + A_1$.

The common root system. The adjoint branching shows that $A_{\bar{\alpha}} \cap \Theta$ contains two full A_2-factors and the A_1 generated by α when $\{\pm\alpha\} \subset \Theta$. Otherwise α is a weight of $(\mathbf{3}, \mathbf{3}, \mathbf{3})$ or its dual and is orthogonal to one root pair in each A_2-factor, giving $A_1 + A_1 + A_1$. The root counts in the table follow. Their strict inequalities give the displayed description of $\mathcal{X}_{\max}$. If α lies in one A_2-factor, a λ satisfying $\langle \lambda, \alpha^\vee \rangle \neq 0$ is nontrivial on that factor and exactly one other factor. The third factor contributes A_2, while the other nontrivial factor contributes A_1. Hence $R_s \cong A_2 + A_1$.

The stabilizer. For $s \in \mathcal{X}_{\max}$, Proposition 4.1 shows that a stabilizer of D_λ fixes λ. A stabilizer of $A_{\bar{\alpha}}$ sends α to $\pm\alpha$, and the nonzero pairing excludes the minus sign. The pair stabilizer therefore fixes $L = \operatorname{span}_{\mathbb{R}}\{\lambda, \alpha\}$ pointwise. Choose $v \in L$ outside the fixed subspaces of Weyl elements that do not fix L pointwise. Then $\operatorname{Stab}_{\mathcal{W}}(v)$ is the pair stabilizer, $\Phi \cap v^\perp = D_\lambda \cap A_{\bar{\alpha}}$, and the Weyl point-stabilizer theorem [13, Ch. V] gives

$$\operatorname{Stab}_{\mathcal{W}}(D_\lambda, A_{\bar{\alpha}}) = W(D_\lambda \cap A_{\bar{\alpha}}) \cong W(A_4).$$

For every $\beta \in R_s$, one has $\beta \perp \lambda$; moreover, either $\beta = \pm\alpha$ or $\beta \perp \alpha$, and $\beta \in \Theta$. The reflection s_β preserves each of D_λ, $A_{\bar{\alpha}}$, and Θ, so $W(R_s)$ fixes the triple. Conversely, every triple stabilizer lies in the pair stabilizer and normalizes R_s; therefore

$$W(R_s) \subseteq \operatorname{Stab}_{\mathcal{W}}(s) \subseteq N_{W(A_4)}(R_s).$$

Inside A_4, the subsystem $A_2 + A_1$ is the $3 + 2$ block decomposition. Its unequal block sizes give $N_{W(A_4)}(R_s) = W(R_s)$, of order 12. Since $|W(E_6)| = 51840$, the orbit of s has 4320 elements.

The orbit count. Affine-diagram symmetry [13, Ch. VI] gives

$$W(A_2)^3 \rtimes S_3 \subseteq N_{\mathcal{W}}(\Theta), \qquad |\mathcal{O}_\Theta| \leq \frac{51840}{6^3 \cdot 6} = 40.$$

Each Θ contains nine root pairs. For a fixed root pair in one A_2-factor, two weights of $\mathbf{3}$ or $\overline{\mathbf{3}}$ pair nontrivially with it; the other nontrivial factor contributes three choices in each of two summands. There are therefore $2 \cdot 3 + 2 \cdot 3 = 12$ compatible weights λ, and $|\mathcal{X}_{\max}| \leq 40 \cdot 9 \cdot 12 = 4320$. The orbit already found attains the bound, proving that the maximizers form one orbit and establishing the stabilizer statement. □

For $\Theta \in \mathcal{O}_\Theta$, write $\Theta = C_1 + C_2 + C_3$ for its three irreducible A_2-components. If $\{\pm\alpha\} \subset \Theta$, let C_α be the component containing α. We say that λ is nontrivial on C_α when $\langle \lambda, \beta^\vee \rangle \neq 0$ for some $\beta \in C_\alpha$.

Theorem 4.3 (Triples with largest common intersection)**.** *The largest possible common intersection has ten roots. The maximizing set is*

$$\mathcal{X}_\cap = \left\{ (D_\lambda, A_{\bar{\alpha}}, \Theta) : \begin{array}{l} \{\pm\alpha\} \subset \Theta, \\ \langle \lambda, \alpha^\vee \rangle = 0, \\ \lambda \text{ is nontrivial on } C_\alpha \end{array} \right\}.$$

For every $s \in \mathcal{X}_\cap$,

$$R_s = D_\lambda \cap A_{\bar{\alpha}} \cap \Theta \cong A_2 + A_1 + A_1.$$

Its stabilizer and orbit size are

$$\operatorname{Stab}_{\mathcal{W}}(s) = W(R_s), \qquad \mathcal{X}_\cap \simeq \mathcal{W}/W(R_s), \qquad |\mathcal{X}_\cap| = 2160.$$

The maximizing triples form a single $\mathcal{W}$-orbit.

Proof. The maximal root count. Suppose first that $\{\pm\alpha\} \not\subset \Theta$. Then $A_{\bar{\alpha}} \cap \Theta \cong A_1^3$, so the common intersection has at most six roots.

Now let $\alpha \in C_1$. The restriction of the **27** to A_2^3 shows that λ is nontrivial on exactly two of the three factors and trivial on the third. If $\langle \lambda, \alpha^\vee \rangle \neq 0$, the factor containing α contributes nothing to the common intersection; the other two factors contribute $A_2 + A_1$, giving eight roots. If the pairing is zero and λ is trivial on C_1, each factor contributes one root pair, giving A_1^3 and six roots. In the remaining case, λ is nontrivial on C_1 but orthogonal to α. The factor on which λ is trivial contributes a full A_2; the other nontrivial factor contributes an A_1; and $\{\pm\alpha\}$ contributes a second A_1. Consequently,

$$R_s \cong A_2 + A_1 + A_1, \qquad |R_s| = 10.$$

These cases prove both the characterization and the maximal value.

The count. Each Θ contains nine root pairs. For a fixed pair in one A_2-factor, consider the two summands of $\mathbf{27}|_{A_2^3}$ that are nontrivial on that factor. In each summand, one of the three weights on the chosen factor is orthogonal to α, while the other nontrivial factor has three choices. This gives $3 + 3 = 6$ weights, and

$$|\mathcal{X}_\cap| = 40 \cdot 9 \cdot 6 = 2160.$$

The stabilizer. By Proposition 4.1, an element stabilizing D_λ fixes λ; it cannot send λ to $-\lambda$, which lies in the dual minuscule orbit. A stabilizer of $A_{\bar{\alpha}}$ sends α to $\pm\alpha$. Put

$$H_0 = \{w \in \mathcal{W} : w\lambda = \lambda,\ w\alpha = \alpha\}.$$

Choose $v \in \mathrm{span}_{\mathbb{R}}\{\lambda, \alpha\}$ outside the fixed subspaces of Weyl elements that do not fix both λ and α. Then $\mathrm{Stab}_{\mathcal{W}}(v) = H_0$, and its orthogonal root system is $A_3 + A_1$. The Weyl point-stabilizer theorem therefore gives $H_0 = W(A_3 + A_1)$. Since $\langle \lambda, \alpha^\vee \rangle = 0$, the reflection s_α fixes λ, commutes with H_0, and exchanges α with $-\alpha$. Therefore

$$\mathrm{Stab}_{\mathcal{W}}(D_\lambda, A_{\bar{\alpha}}) = H_0 \times \langle s_\alpha \rangle = W(A_3 + A_1 + A_1).$$

For every $\beta \in R_s$, one has $\beta \perp \lambda$; moreover, either $\beta = \pm\alpha$ or $\beta \perp \alpha$, and $\beta \in \Theta$. Thus s_β preserves each member of the triple, so $W(R_s) \subseteq \mathrm{Stab}_{\mathcal{W}}(s)$. Conversely, a triple stabilizer lies in the pair stabilizer and normalizes R_s. Inside $W(A_3 + A_1 + A_1)$, the subsystem R_s contains both A_1-components and an A_2 in the A_3-factor. In $W(A_3) \cong S_4$, the corresponding $S_3 = W(A_2)$ is self-normalizing, so

$$W(R_s) \subseteq \mathrm{Stab}_{\mathcal{W}}(s) \subseteq N_{W(A_3+A_1+A_1)}(R_s) = W(R_s).$$

The stabilizer has order 24, so its orbit has $51840/24 = 2160$ elements. This is the number of maximizers, and the maximizing set is one orbit. □

Since $|D_\lambda \cap \Theta| = 10$ for every pair (D_λ, Θ), the common-maximum condition has the equivalent form

$$|D_\lambda \cap A_{\bar{\alpha}} \cap \Theta| = 10 \quad \Longleftrightarrow \quad D_\lambda \cap \Theta \subseteq A_{\bar{\alpha}}.$$

The common maximum occurs when the third subsystem contains the rigid ten-root intersection of the other two.

The two extremal criteria are inequivalent. Pairwise maximality gives a common intersection of type $A_2 + A_1$ with eight roots, whereas maximizing the common intersection gives type $A_2 + A_1 + A_1$ with ten roots.

Remark (Monotone objectives). Any function of $(|D \cap A|, |D \cap \Theta|, |A \cap \Theta|)$ that is strictly increasing in each coordinate is maximized on the pairwise-maximal orbit. Likewise, every strictly increasing function of $|D \cap A \cap \Theta|$ is maximized on the common-maximal orbit.

4.3 The six orbit cells and their chiral parents

A triple offers three canonical pair-first parents. Their chiral dimensions first distinguish the parent-level one-family criterion specified below; for that extraction, the common root systems then show that color occurs on exactly the two extremal cells.

Proposition 4.4 (Chiral dimensions of the three pair-first parents)**.** *For a representation $R = \bigoplus_U m_U U$ of a semisimple compact group, define its chiral dimension by*

$$d_\chi(R) = \sum_{\{U,U^*\}} |m_U - m_{U^*}| \, \dim U,$$

where the sum runs over unordered pairs of non-self-dual irreducibles. For $P, Q \in \{D_\lambda, A_{\bar{\alpha}}, \Theta\}$, let

$$H_{PQ} = [(\widehat{P} \cap \widehat{Q})^\circ, (\widehat{P} \cap \widehat{Q})^\circ].$$

Then the restriction of the **27** *has*

parent	*root type*	$d_\chi(\mathbf{27})$
H_{DA}	A_4 *or* $A_3 + A_1 + A_1$	15 *or* 16
$H_{D\Theta}$	$A_2 + A_1 + A_1$	12
$H_{A\Theta}$	$A_2 + A_2 + A_1$ *or* A_1^3	27 *or* 0.

The three orderings are therefore distinguished by their chiral content: H_{DA} gives 15 *or* 16*, $H_{D\Theta}$ gives* 12*, and $H_{A\Theta}$ gives* 27 *or* 0*. The first two values in the H_{DA} row are the dimensions of the familiar one-family $SU(5)$ and* Spin(10)*/Pati–Salam chiral multiplets.*

Proof. For the two *D*-*A* strata, the familiar restrictions are

$$\mathbf{27}|_{A_4} = \mathbf{10} \oplus 2\,\overline{\mathbf{5}} \oplus \mathbf{5} \oplus 2\,\mathbf{1}$$

and

$$\mathbf{27}|_{A_3+A_1+A_1} = (\mathbf{4},\mathbf{2},\mathbf{1}) \oplus (\overline{\mathbf{4}},\mathbf{1},\mathbf{2}) \oplus (\mathbf{6},\mathbf{1},\mathbf{1}) \oplus (\mathbf{1},\mathbf{2},\mathbf{2}) \oplus (\mathbf{1},\mathbf{1},\mathbf{1}),$$

which give chiral dimensions 15 and 16.

For the other two orderings, use

$$\mathbf{27}|_{A_2^3} = (\mathbf{3},\overline{\mathbf{3}},\mathbf{1}) \oplus (\mathbf{1},\mathbf{3},\overline{\mathbf{3}}) \oplus (\overline{\mathbf{3}},\mathbf{1},\mathbf{3}).$$

The subgroup $H_{D\Theta}$ contains one full A_2 factor and one A_1 in each of the other two. In the order $A_1 \times A_1 \times A_2$,

$$\begin{aligned}\mathbf{27} = {} & (\mathbf{2},\mathbf{2},\mathbf{1}) \oplus (\mathbf{2},\mathbf{1},\mathbf{1}) \oplus (\mathbf{1},\mathbf{2},\mathbf{1}) \oplus (\mathbf{1},\mathbf{1},\mathbf{1}) \\ & \oplus (\mathbf{1},\mathbf{2},\overline{\mathbf{3}}) \oplus (\mathbf{1},\mathbf{1},\overline{\mathbf{3}}) \oplus (\mathbf{2},\mathbf{1},\mathbf{3}) \oplus (\mathbf{1},\mathbf{1},\mathbf{3}).\end{aligned}$$

The first line is self-dual, and the two three-dimensional terms in the second line form a dual pair. The remaining two summands have dimension six, so $d_\chi = 12$.

If $H_{A\Theta} \cong A_1 + A_2 + A_2$, then

$$\mathbf{27} = (\mathbf{2},\overline{\mathbf{3}},\mathbf{1}) \oplus (\mathbf{1},\overline{\mathbf{3}},\mathbf{1}) \oplus (\mathbf{1},\mathbf{3},\overline{\mathbf{3}}) \oplus (\mathbf{2},\mathbf{1},\mathbf{3}) \oplus (\mathbf{1},\mathbf{1},\mathbf{3}),$$

and none of the dual summands occurs, so $d_\chi = 27$. If $H_{A\Theta} \cong A_1^3$, every irreducible is self-dual and $d_\chi = 0$. □

Corollary 4.5 (Uniqueness of the one-family pair-first parent)**.** *Among the three pair-first constructions, H_{DA} is the unique one whose parent-level chiral dimension is* 15 *or* 16*. Define the one-family parent criterion to require that the non-self-dual remainder of one* **27***, computed as an H_{PQ}-module, have dimension* 15 *or* 16*. Under interpretation* (M) *of Section 5 and this criterion, the D-A parent is uniquely characterized. The criterion is evaluated before further restriction; the Standard Model representations and charges are identified by the subgroup chain and branching in Section 5.*

Proposition 4.6 (The color-containing orbits)**.** *For $s = (D_\lambda, A_{\bar\alpha}, \Theta) \in \mathcal{X}$, put*

$$R_s = D_\lambda \cap A_{\bar\alpha} \cap \Theta.$$

Then R_s contains an irreducible component of type A_2 if and only if s is pairwise maximal or has largest common intersection.

More generally, define

$$S_s = [(\widehat{D}_\lambda \cap \widehat{A}_{\bar\alpha})^\circ, (\widehat{D}_\lambda \cap \widehat{A}_{\bar\alpha})^\circ], \qquad K_s = (S_s \cap \widehat{\Theta})^\circ.$$

The semisimple root system of K_s is R_s. Among all 38,880 *triples, the extracted group K_s contains a simple $\mathfrak{su}(3)$ factor only on the two extremal classes.*

Proof. Write $\Theta = C_1 + C_2 + C_3$. If $\{\pm\alpha\} \not\subset \Theta$, then $A_{\bar\alpha} \cap \Theta \cong A_1^3$, so every component of R_s has type A_1. Now suppose $\alpha \in C_1$. The restriction of the **27** to A_2^3 shows that λ is nontrivial on exactly two factors. If $\langle \lambda, \alpha^\vee \rangle \neq 0$, the proof of Theorem 4.2 gives $R_s \cong A_2 + A_1$, and the triple is pairwise maximal. If the pairing vanishes and λ is nontrivial on C_1, the proof of Theorem 4.3 gives $R_s \cong A_2 + A_1 + A_1$, and the common intersection is maximal. In the remaining case λ is trivial on C_1, and $R_s \cong A_1^3$. An A_2-component occurs only in the two extremal cases.

The roots of S_s are $D_\lambda \cap A_{\bar\alpha}$. Since $\widehat{\Theta}$ contains the fixed maximal torus, the roots of the derived algebra of K_s are exactly their intersection with Θ, namely R_s, which proves the group statement. □

Since $|\mathcal{O}_D| = 27$, $|\mathcal{O}_A| = 36$, and $|\mathcal{O}_\Theta| = 40$,

$$|\mathcal{X}| = 27 \cdot 36 \cdot 40 = 38{,}880.$$

Table 2 lists the six incidence cells. The independent scan also shows that each row is a Weyl orbit.

Table 2: The six Weyl orbits in the triple space.

pairwise profile	$\lvert R_s\rvert$	type of R_s	status	triples
$(20, 10, 14)$	8	$A_2 + A_1$	pairwise maximum	4,320
$(16, 10, 14)$	10	$A_2 + A_1 + A_1$	common maximum	2,160
$(20, 10, 6)$	4	$A_1 + A_1$	nonextremal	12,960
$(16, 10, 14)$	6	$A_1 + A_1 + A_1$	nonextremal	3,240
$(16, 10, 6)$	6	$A_1 + A_1 + A_1$	nonextremal	3,240
$(16, 10, 6)$	2	A_1	nonextremal	12,960

Remark (Full root-system automorphisms). Let τ be the nontrivial diagram automorphism of E_6, and let w_0 be the longest Weyl element. Since $w_0\tau = -1$ on the root system and every closed root subsystem is stable under negation, τ acts on closed subsystems exactly as w_0^{-1}. Hence the six $W(E_6)$-orbits above are already the six orbits under $\mathrm{Aut}\,\Phi(E_6)$; no cells merge when the full root-system automorphism group is used, and each stabilizer doubles in order. On representations the outer class exchanges **27** and $\overline{\mathbf{27}}$, but that orientation exchange leaves the unoriented subsystem classification unchanged.

Appendix B reconstructs the root system and Weyl group from the Cartan matrix, scans all 38,880 triples, and verifies the two extremal orbit theorems and the six-cell classification.

4.4 Faithful subgroup chains

The common root systems record the derived semisimple parts of the connected intersections, not their full tori. The abelian directions are recovered from the ambient maximal torus after the indicated commutator step. In the pairwise-maximal case the root system $A_2 + A_1$, together with one surviving circle, gives the rank-four group $S(U(3) \times U(2))$; the faithful quotient is determined below.

The root intersections determine the local semisimple types. The next result identifies their faithful compact global forms and compares the two chains.

Remark (Fixed groups and the intersection construction). The regular groups associated with D_5, $A_5 + A_1$, and A_2^3 are the connected fixed groups of two toral involutions and of the order-three grading. Their root systems are reconstructed directly in Appendices A and B. The defining toral elements, and hence their inner automorphisms, may be chosen in the same maximal torus and therefore commute. Their fixed groups need not commute; the construction uses their intersections. Intersections such as $G_{\mathrm{SM}} = SU(5) \cap G_{\mathrm{PS}}$ inside Spin(10) are familiar from unified and orbifold constructions [17]. Theorems 4.2 and 4.3 add a classification statement: the two chains arise from the two extremal Weyl orbits in the full triple space, with their faithful global forms and compatible $\mathbb{Z}_6$ kernels.

Corollary 4.7 (The Standard Model and left-right subgroup chains)**.** *For a pairwise maximizer $s_P = (D, A_P, \Theta)$, define*

$$L = [\widehat{D}, \widehat{D}], \qquad S_P = [(\widehat{D} \cap \widehat{A}_P)^\circ, (\widehat{D} \cap \widehat{A}_P)^\circ], \qquad K_P = (S_P \cap \widehat{\Theta})^\circ.$$

Then

$$K_P \cong G_{\mathrm{SM}} = S(U(3) \times U(2)) \subset S_P \cong SU(5) \subset L \cong \mathrm{Spin}(10).$$

Its global form is

$$G_{\mathrm{SM}} \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

For a triple $s_T = (D, A_T, \Theta)$ attaining the common maximum, define

$$S_T = [(\widehat{D} \cap \widehat{A}_T)^\circ, (\widehat{D} \cap \widehat{A}_T)^\circ], \qquad K_T = (S_T \cap \widehat{\Theta})^\circ.$$

Then

$$K_T \cong G_{\mathrm{LR}} \subset S_T \cong G_{\mathrm{PS}} \subset \mathrm{Spin}(10),$$

where the faithful Pati–Salam and left-right groups are

$$G_{\mathrm{PS}} \cong \frac{SU(4) \times SU(2)_L \times SU(2)_R}{\langle(-I_4, -I_2, -I_2)\rangle}, \qquad G_{\mathrm{LR}} \cong \frac{SU(3) \times SU(2)_L \times SU(2)_R \times U(1)_{B-L}}{\mathbb{Z}_6}.$$

If s_P and s_T have the same D and Θ, then

$$G_{\mathrm{SM}} \subset G_{\mathrm{LR}} \subset G_{\mathrm{PS}} \subset \mathrm{Spin}(10) \subset E_6^{\mathrm{sc}}, \qquad Y = T_R^3 + \frac{B - L}{2}.$$

There are 8,640 *such ordered pairs (s_P, s_T).*

Proof. The pairwise-maximal chain. For s_P, the root systems of L, S_P, K_P are $D_5, A_4, A_2 + A_1$. Pullback of the **27** to Spin(10) contains the faithful half-spinor **16**; pullback to $SU(5)$ contains a faithful **5** or $\overline{\mathbf{5}}$. Hence $L \cong \mathrm{Spin}(10)$ and $S_P \cong SU(5)$. The connected full-rank subgroup of $SU(5)$ with roots $A_2 + A_1$ is $S(U(3) \times U(2))$, with covering map

$$(g_3, g_2, z) \longmapsto \mathrm{diag}(z^{-2} g_3, z^3 g_2)$$

and kernel $\mathbb{Z}_6$.

The common-maximal chain. For s_T, the root systems of S_T and K_T are $A_3 + A_1 + A_1$ and $A_2 + A_1 + A_1$. Under the universal cover $SU(4) \times SU(2)_L \times SU(2)_R$, the **27** contains

$$(\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\overline{\mathbf{4}}, \mathbf{1}, \mathbf{2}) \oplus (\mathbf{6}, \mathbf{1}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2}, \mathbf{2}) \oplus (\mathbf{1}, \mathbf{1}, \mathbf{1}).$$

The common kernel on these summands is generated by $(-I_4, -I_2, -I_2)$, so its faithful image in Spin(10) is

$$S_T \cong G_{\mathrm{PS}} \cong \frac{SU(4) \times SU(2)_L \times SU(2)_R}{\mathbb{Z}_2}.$$

Inside $SU(4)$, the full-rank subgroup with roots A_2 is $S(U(3) \times U(1))$. Accordingly, K_T is the image of

$$\begin{aligned} SU(3) \times SU(2)_L \times SU(2)_R \times U(1) &\longrightarrow S_T, \\ (g_3, g_L, g_R, z) &\longmapsto [\mathrm{diag}(z g_3, z^{-3}), g_L, g_R]. \end{aligned}$$

Its kernel is cyclic of order six, generated by

$$\left(e^{2\pi i/3} I_3, -I_2, -I_2, e^{\pi i/3}\right).$$

The $U(1)$ generator in $SU(4)$ is proportional to $\mathrm{diag}(1, 1, 1, -3)$, which is the usual $B - L$ generator, giving the stated global form of G_{LR} [19, 14, 15].

Containment of the root systems. Fix (D_λ, Θ), and write $\Theta = C_1 + C_2 + C_3$ so that λ is nontrivial on C_1, C_2 and trivial on C_3. For $i = 1, 2$, let $\{\pm\rho_i\}$ be the unique root pair in C_i orthogonal to λ. A pairwise maximizer chooses one of the other two root pairs in C_1 or C_2, giving four choices for A_P. Its common root system is

$$C_3 + \{\pm\rho_2\} \quad \text{or} \quad C_3 + \{\pm\rho_1\},$$

according to the factor containing the chosen root. A triple attaining the common maximum chooses $\{\pm\rho_1\}$ or $\{\pm\rho_2\}$, giving two choices for A_T, and in either case its common root system is

$$C_3 + \{\pm\rho_1\} + \{\pm\rho_2\}.$$

For fixed (D_λ, Θ), the common root subsystem is therefore independent of the two admissible choices of A_T. The root system of K_P is contained in that of K_T for all eight pairings.

The Cartan subalgebras are compatible as well. The roots of S_P span a four-dimensional subspace of $\lambda^\perp$, while the roots of S_T have rank five and span $\lambda^\perp$. Since $\widehat{\Theta}$ contains T_0, the Cartan subalgebras of K_P and K_T are those of S_P and S_T, respectively, so $K_P \subset K_T$.

Compatibility of the global forms. Put $\omega = e^{2\pi i/3}$ and $\zeta = e^{\pi i/3}$. The covering kernels of G_{SM} and G_{LR} are generated, respectively, by

$$k_{\mathrm{SM}} = (\omega I_3, -I_2, \zeta), \qquad k_{\mathrm{LR}} = (\omega I_3, -I_2, -I_2, \zeta).$$

With consistent orientations of the two abelian factors, the inclusion lifts to

$$\widetilde{\iota}: SU(3)\times SU(2)_L\times U(1)\longrightarrow SU(3)\times SU(2)_L\times SU(2)_R\times U(1),$$
$$(g_3,g_L,u)\longmapsto\left(g_3,g_L,\begin{pmatrix}u^3&0\\0&u^{-3}\end{pmatrix},u\right).$$

It sends k_{SM} to k_{LR}, so it descends to $G_{\mathrm{SM}}\hookrightarrow G_{\mathrm{LR}}$. On Lie algebras its abelian generator is

$$6Y=6T_R^3+3(B-L),\qquad\text{equivalently}\qquad Y=T_R^3+\frac{B-L}{2}.$$

The containment count. There are $27\cdot 40=1080$ choices of (D,Θ). Each has four pairwise-maximal and two common-maximal choices of A, so there are $1080\cdot 4\cdot 2=8640$ ordered pairs (s_P,s_T). □

Remark (Covering groups and faithful quotients). Writing tildes for the displayed covering products, the covering-product and quotient maps form the commutative diagram

$$\begin{array}{ccccc}\widetilde{G}_{\mathrm{SM}}&\hookrightarrow&\widetilde{G}_{\mathrm{LR}}&\xrightarrow{\pi}&\widetilde{G}_{\mathrm{PS}}\\\downarrow&&\downarrow&&\downarrow\\G_{\mathrm{SM}}&\hookrightarrow&G_{\mathrm{LR}}&\hookrightarrow&G_{\mathrm{PS}},\end{array}$$

where the three vertical kernels are $\mathbb{Z}_6$, $\mathbb{Z}_6$, and $\mathbb{Z}_2$, respectively. The first upper arrow is the injective lift $\widetilde{\iota}$ displayed above. The second is the natural covering-product map and has

$$\ker\pi=\langle k_{\mathrm{LR}}^2\rangle\cong\mathbb{Z}_3.$$

It sends k_{LR} to the generator of the Pati–Salam $\mathbb{Z}_2$ kernel. Thus its full inverse image is $\langle k_{\mathrm{LR}}\rangle$, and quotienting gives the injective lower arrow $G_{\mathrm{LR}}\hookrightarrow G_{\mathrm{PS}}$. The first lifted inclusion sends the Standard Model kernel generator to k_{LR}, so the charge lattice is compatible along the chain.

Remark (Interchange symmetry over a fixed (D,Θ)). The 1,080 pairs (D_λ,Θ) form one Weyl orbit. By the injectivity of $\lambda\mapsto D_\lambda$, every element stabilizing the pair fixes λ, so the context stabilizer lies in $W(D_\lambda)\cong W(D_5)$. If $R=D_\lambda\cap\Theta\cong A_2+A_1+A_1$, a direct normalizer calculation gives

$$\mathrm{Stab}_{\mathcal{W}}(D_\lambda,\Theta)=N_{W(D_5)}(R)\cong W(R)\rtimes\mathbb{Z}_2,\qquad|\,\mathrm{Stab}\,|=48.$$

The nontrivial factor exchanges the two A_1-components, equivalently the two distinguished root pairs $\{\pm\rho_1\}$ and $\{\pm\rho_2\}$. It fixes the common-maximal root system

$$C_3+\{\pm\rho_1\}+\{\pm\rho_2\}$$

and exchanges the two pairwise-maximal root systems $C_3+\{\pm\rho_1\}$ and $C_3+\{\pm\rho_2\}$. This is an internal Weyl symmetry of the subgroup configuration, distinct from any spacetime parity operation.

5 The 27 and the Standard Model

No further selection principle enters here. We restrict the already selected module along the faithful subgroup chain, retain the complete branching, and then pass to its net chiral class. The decomposition and chiral index are representation-theoretic statements independent of the interpretation below.

Complete-matter interpretation (M). To identify the chiral index with the complete family multiplicity, we assume that one complex isotropy half is the full chiral fermion module. The homogeneous uniqueness theorem, the subgroup classifications, and the branching calculation do not depend on (M).

5.1 The faithful Standard Model group and its charge lattice

Corollary 4.7 gives one E_8-conjugacy class, so the restriction may be computed in the block representative

$$G_{\mathrm{SM}} = S(U(3)\times U(2)) \subset SU(5) \subset \mathrm{Spin}(10) \subset E_6^{\mathrm{sc}}.$$

A primitive cocharacter generating its central $U(1)$ is, up to inversion,

$$\eta(z) = \mathrm{diag}(z^{-2}I_3, z^3 I_2).$$

Orient η so that the $(\mathbf{3},\mathbf{2})$ in the $\mathbf{10}$ has integer charge $q=1$; conventional hypercharge is then $Y = q/6$. Define $t \in \mathbb{Z}/3$ and $s \in \mathbb{Z}/2$ by requiring the central elements $e^{2\pi i/3}I_3 \in SU(3)$ and $-I_2 \in SU(2)$ to act as $e^{2\pi i t/3}$ and $(-1)^s$, respectively. The kernel in Corollary 4.7 says that a representation of integer charge q descends to G_{SM} exactly when

$$q + 2t + 3s \equiv 0 \pmod 6.$$

This congruence is the $\mathbb{Z}_6$ charge lattice. When $t=0$, it links q to the $SU(2)$ center parity and forces every electric charge in a color-singlet multiplet to be integral.

5.2 One 27: a chiral family, a vectorlike pair, and two singlets

Restriction along this subgroup chain gives

$$\begin{aligned} \mathbf{27} \downarrow \mathrm{Spin}(10) &= \mathbf{16} \oplus \mathbf{10} \oplus \mathbf{1} \\ \downarrow SU(5) &= \underbrace{(\mathbf{10} \oplus \overline{\mathbf{5}})}_{\mathcal{F}_{\mathrm{SM}}} \oplus (\mathbf{5} \oplus \overline{\mathbf{5}}) \oplus 2\,\mathbf{1}. \end{aligned} \tag{8}$$

The restriction $\mathcal{F}_{\mathrm{SM}} = (\mathbf{10}\oplus\overline{\mathbf{5}})|_{G_{\mathrm{SM}}}$ is $(\mathbf{3},\mathbf{2})_{1/6} \oplus (\overline{\mathbf{3}},\mathbf{1})_{-2/3} \oplus (\mathbf{1},\mathbf{1})_1 \oplus (\overline{\mathbf{3}},\mathbf{1})_{1/3} \oplus (\mathbf{1},\mathbf{2})_{-1/2}$, the fermion representation of one minimal Standard Model family in an all-left-handed convention [14, 15]. The G_{SM}-singlet $\nu^c \subset \mathbf{16}$ is one of the two singlets in (8), not a summand of $\mathcal{F}_{\mathrm{SM}}$. Equations (7)–(8) give the full decomposition

$$V|_{G_{\mathrm{SM}}} \cong 3\,\mathcal{F}_{\mathrm{SM}} \oplus 3(\mathbf{5}|_{G_{\mathrm{SM}}} \oplus \overline{\mathbf{5}}|_{G_{\mathrm{SM}}}) \oplus 6\,\mathbf{1}. \tag{9}$$

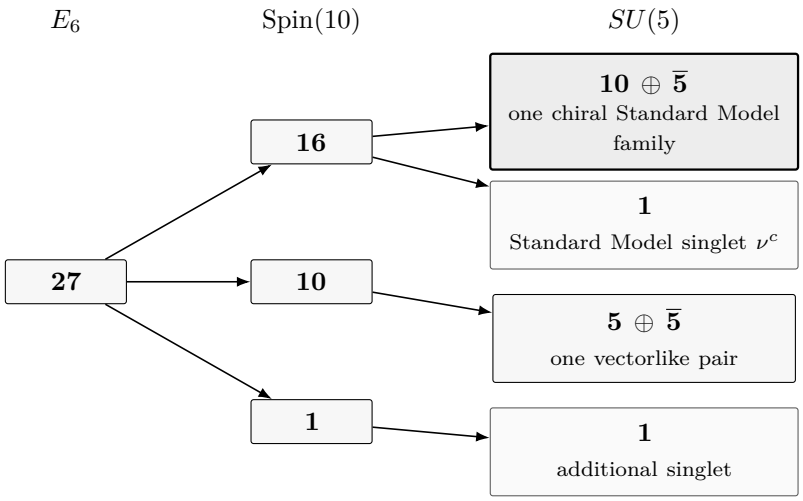


Figure 2: One **27** branches into a chiral Standard Model family, a vectorlike $\mathbf{5} \oplus \overline{\mathbf{5}}$ pair, and two singlets. The factor $\mathbf{3}_M$ gives three copies of this branching.

The six singlets consist of three ν^c-type states and three further neutral states. This is the field content used in extended-seesaw constructions [18]; the classification fixes the representations, not their masses or couplings.

5.3 The net chiral index is three

Let $R(G_{\mathrm{SM}})$ be the additive Grothendieck group of finite-dimensional complex representations. Let χ be the quotient map obtained by imposing $\chi(R^*) = -\chi(R)$ and $\chi(S) = 0$ for every self-dual S. The map χ records, for each non-self-dual pair $\{R, R^*\}$, the multiplicity difference $m_R - m_{R^*}$. Applying it to (9) gives

$$\begin{aligned} \chi(V|_{G_{\mathrm{SM}}}) &= 3\chi(\mathcal{F}_{\mathrm{SM}}) + 3(\chi(\mathbf{5}|_{G_{\mathrm{SM}}}) + \chi(\overline{\mathbf{5}}|_{G_{\mathrm{SM}}})) + 6\chi(\mathbf{1}) \\ &= \boxed{3\chi(\mathcal{F}_{\mathrm{SM}})}. \end{aligned} \tag{10}$$

The vectorlike pairs and singlets make no contribution. Equation (10) records net chirality; equation (9) records the full module. Duality gives

$$\{\chi(V|_{G_{\mathrm{SM}}}), \chi(V^*|_{G_{\mathrm{SM}}})\} = \{3\chi(\mathcal{F}_{\mathrm{SM}}), -3\chi(\mathcal{F}_{\mathrm{SM}})\}.$$

Relative to $\chi(\mathcal{F}_{\mathrm{SM}})$, the coefficients $+3$ and -3 are the chiral indices of V and V^*, respectively.

Corollary 5.1 (Chiral content of the classified module)**.** *The two orientations have chiral-index coefficients $+3$ and -3, relative to $\chi(\mathcal{F}_{\mathrm{SM}})$. Beyond this family class, no additional net chiral class occurs in V or V^*; every remaining summand is vectorlike or a singlet. Under the complete-matter interpretation* (M)*, the absolute coefficient 3 is the complete physical family multiplicity.*

5.4 Local anomalies and the $SU(3)_M^3$ background anomaly

Standard Model insertions. Perturbative local cubic and linear anomaly coefficients are additive, odd under duality, and zero on self-dual modules, so they factor through χ. The linear coefficient governs the mixed gauge–gravitational anomaly. Since $\mathrm{Lie}(G_{\mathrm{SM}}) \subset \mathfrak{e}_6 = \mathrm{Rad}(c_V)$, the cubic anomaly

of $V|_{G_{\mathrm{SM}}}$ vanishes. The algebra $\mathfrak{e}_6$ is perfect, so every finite-dimensional trace functional on it also vanishes. Equation (10) then shows that one copy of $\mathcal{F}_{\mathrm{SM}}$, and hence three copies, is free of perturbative gauge and mixed gauge–gravitational anomalies.

The preceding argument covers perturbative local anomalies. The spectrum in (9) also has no mod-two $SU(2)$ anomaly: counting color multiplicity, each $\mathcal{F}_{\mathrm{SM}}$ contains four doublets, while each vectorlike $\mathbf{5} \oplus \overline{\mathbf{5}}$ contributes two. The same spectrum is the restriction of anomaly-free $SU(5)$ representations. Bordism analyses of the four connected Standard Model global forms show that this embedding has no further global gauge anomaly for the $\mathbb{Z}_6$ quotient used here [29].

The family background. To identify the remaining cubic anomaly, let F_{SM} be a G_{SM}-gauge field strength and F_M a background field strength for $SU(3)_M$. On $V = \mathbf{27} \otimes \mathbf{3}_M$, put

$$F = F_{\mathrm{SM}} \otimes 1_{\mathbf{3}} + 1_{\mathbf{27}} \otimes F_M.$$

The radical condition kills every cubic term containing F_{SM}, and both simple factors act tracelessly. Therefore

$$\mathrm{Tr}_V(F^3) = 27\ \mathrm{Tr}_{\mathbf{3}}(F_M^3), \qquad \mathrm{Tr}_V(F) = 0. \tag{11}$$

Every perturbative local anomaly coefficient with a G_{SM} gauge insertion vanishes, including coefficients mixed with the background $SU(3)_M$ field. The only remaining cubic component is the $SU(3)_M^3$ term, with coefficient 27 when $A(\mathbf{3}) = 1$. As a background global symmetry, the commuting $SU(3)_M$ carries this perturbative $SU(3)_M^3$ anomaly. Promoting it to a gauge factor requires additional anomaly-cancelling structure.

6 Discussion

Physical status of the domain. The hypotheses do not name the answer. Compactness is the standard finite-dimensional setting for unitary internal symmetry; effectivity removes generators that act on nothing; complex type supplies the representation-theoretic realization of gauge chirality used here; and perturbative cubic anomaly cancellation is required on gauged directions. Irreducibility and the ideal-valued radical impose the stricter claims that the homogeneous matter sector is one structure and that cubic-anomaly safety is found inside that structure rather than supplied by a separately chosen subgroup. The unifying premise is that the matter module is canonical to an embedding, not selected after the internal algebra. That premise has a long history in coset reduction and homogeneous family unification; the exhaustive converse classification is the new result.

The theorem is conditional in the ordinary mathematical sense. Experiment decides whether nature realizes this homogeneous encoding; the classification decides what the encoding entails. The two questions are complementary. The homogeneous theorem has the stated domain and equivalence. The subgroup results additionally use the maximal-subsystem triple construction, the parent-level and extremal criteria, and the specified derived-subgroup operations. Interpretation (M) identifies the resulting chiral coefficient with the complete physical family multiplicity.

Uniqueness and no-go. The forward statement is Theorem 2.1. Its contrapositive is the no-go statement: any connected, infinitesimally effective compact homogeneous realization with irreducible complex-type matter that is not the pair of Theorem 2.1 has zero cubic-anomaly radical. If nature admits the domain, the internal pair has one form; if observations require an incompatible internal structure, at least one defining premise fails.

The later conclusions have their own, narrower contrapositives. Additional net chiral matter incompatible with the complete branching of Section 5 would challenge (M) or the homogeneous identification, not the Lie-theoretic uniqueness theorem by itself. A different mechanism for the low-energy subgroup would challenge the maximal-subsystem triple realization, not the unique homogeneous pair. These distinctions turn the theorem into a diagnostic boundary rather than an elastic model.

Representation theory and compression. There is nothing mysterious about Lie theory appearing in particle physics: continuous symmetry leads to Lie groups, quantum multiplets to their representations, branching to roots and weights, and faithful charge lattices to weight and cocharacter lattices [27]. The surprise is the amount of compression. The premises name none of E_8, E_6, $SU(3)$, **27**, the multiplicity three, the Standard Model group, or its $\mathbb{Z}_6$ quotient. What appear in conventional model building as separate choices—group, representation, cubic-anomaly-safe sector, replication, breaking chain, and global form—collapse here into one homogeneous theorem, one finite incidence theorem, and one restriction calculation. A unique answer is natural once a domain has been closed; the unexpected fact is how little answer-specific information is needed to close this one.

7 Conclusion

The compact homogeneous classification has one solution, up to isomorphism of Lie-algebra pairs and duality:

$$(\mathfrak{e}_8, \mathfrak{e}_6 \oplus \mathfrak{su}(3)_M, \{\mathbf{27} \boxtimes \mathbf{3}, \overline{\mathbf{27}} \boxtimes \overline{\mathbf{3}}\}).$$

Its cubic radical is $\mathfrak{e}_6$. Every other connected, infinitesimally effective compact pair with irreducible complex-type isotropy has zero radical.

For the selected E_6, the complete maximal-subsystem triple space has 38,880 elements and six Weyl orbits. Exactly two have a common A_2 component. Compatible representatives give

$$G_{\mathrm{SM}} \subset G_{\mathrm{LR}} \subset G_{\mathrm{PS}} \subset \mathrm{Spin}(10) \subset E_6^{\mathrm{sc}}, \qquad Y = T_R^3 + \frac{B-L}{2},$$

with faithful global forms, compatible $\mathbb{Z}_6$ kernels, and 8,640 labelled containment incidences representing 2,160 distinct root-subsystem inclusions.

Along this Standard Model chain,

$$V|_{G_{\mathrm{SM}}} \cong 3\,\mathcal{F}_{\mathrm{SM}} \oplus 3\,(\mathbf{5} \oplus \overline{\mathbf{5}})|_{G_{\mathrm{SM}}} \oplus 6\,\mathbf{1}.$$

The two orientations have chiral indices $+3$ and -3; all other summands are vectorlike or singlets. Under (M), the coefficient is the complete family multiplicity of the module.

The theorem is both a conditional derivation and a no-go result. If the compact homogeneous domain describes the internal structure of nature, it admits one solution; if nature requires a different structure, at least one defining premise fails. Within the stated domains, the internal algebra, matter module, cubic-anomaly-safe ideal, three-dimensional multiplicity factor, two color-containing chains, faithful charge lattice, and net chiral index are not a list of independent model-building choices. They are one classified structure.

A Direct verification of the homogeneous census

Three exact programs reproduce the finite parts of Section 3 from Cartan matrices and Weyl's dimension formula. They use only integer and rational arithmetic.

A.1 Toral gradings

The script `check_toral_gradings.py` generates the exceptional root systems by reflection closure and enumerates every nonzero functional on the root lattice modulo two and modulo three, up to the Weyl action. It retains the gradings whose grade-zero roots span the full rank. In the involutive case every semisimple row has a connected odd root graph; compact conjugation preserves that graph, so its module is self-conjugate. In the order-three case, write Φ_1 for the roots of degree one. The irreducible rows are

$\mathfrak{g}$	$\mathfrak{g}_0^{\mathrm{ss}}$	$\lvert\Phi_1\rvert = \dim \mathfrak{g}_1$
G_2	A_2	3
F_4	$A_2 + A_2$	18
E_6	A_2^3	27
E_7	$A_5 + A_2$	45
E_8	A_8	84
E_8	$E_6 + A_2$	81.

The two coordinate representatives in type E_7 lie in one Weyl orbit modulo three. For the classical types the corresponding centralizers are computed directly in the defining module, as in the proof of Proposition 3.3.

A.2 Exceptional rank-deficient sieve

The script `check_rank_deficient_exceptional.py` implements Proposition 3.5. It first verifies $T(\mathrm{ad}) = h^\vee$ for every simple type and the positive controls

$$T_{A_2}(\mathbf{3}) = \tfrac{1}{2}, \quad T_{B_3}(\mathbf{7}) = 1, \quad T_{C_3}(\mathbf{6}) = \tfrac{1}{2}, \quad T_{G_2}(\mathbf{7}) = 1, \quad T_{F_4}(\mathbf{26}) = 3, \quad T_{E_6}(\mathbf{27}) = 3.$$

For each exceptional ambient algebra it then enumerates every multiset of compact simple factors of smaller total rank and every irreducible factor module inside the separate bounds (6b). Tensor products are tested against

$$\dim \mathfrak{g} = \sum_i \dim \mathfrak{h}_i + 2\prod_i d_i, \qquad h_{\mathfrak{g}}^\vee I_i = h_i^\vee + 2T_i \prod_{j\neq i} d_j,$$

complex type, and the vanishing of at least one simple-factor cubic. Counts use labelled Cartan-type presentations, including both B_2 and C_2; permutations of identical labels are identified. This redundancy has no effect on the empty final necessary locus. The successive counts are

	factor multisets	dimension	complex	radical	indices
G_2	0	0	0	0	0
F_4	5	0	0	0	0
E_6	28	19	12	10	0
E_7	48	15	2	2	0
E_8	118	30	20	14	0.

The file `exceptional_rank_deficient_audit.json` records every representation product reaching the index stage and the nonintegral index that excludes it. Dimension and index arithmetic are not used as an embedding classification: this is a no-go sieve, and an empty necessary locus is sufficient for the proposition.

A.3 Classical highest-weight arithmetic

The script `check_rank_deficient_classical.py` verifies the Dynkin-label equations and the arithmetic used in Proposition 3.7. In particular, it checks the only solutions of

$$\lambda + \lambda^* = \theta, \qquad 2\lambda = \theta, \qquad 2\lambda - \alpha_i = \theta,$$

the identity, with $1 \le i < N/2$ and $a \ge 1$,

$$C_3(2a(\omega_i + \omega_{N-i}) - \alpha_i) = -6a(2a + N - 2i),$$

and the two odd-D_r dimension ratios displayed in the proof. It also checks symbolically, by polynomial coefficient arithmetic, the monotonicity inequalities used for all odd $r \ge 5$, together with $D_{r,1} = \binom{2r}{r-1}$. The four remaining E_6 dimension gaps are reconstructed directly from its Cartan matrix. The Cartan-equation regression runs through rank 32, and the direct A-type cubic substitution runs through $N = 32$ and $1 \le a \le 5$; these bounded checks audit formulas used by the manuscript's all-rank argument rather than replacing that argument. The odd-D_r monotonicity check is genuinely symbolic and supplies its stated all-rank polynomial inequalities.

B Exact verification of the E_6 incidence classification

The finite statements in Section 4 admit a complete independent reconstruction from the E_6 Cartan matrix. Two implementations use different root-system realizations and Weyl actions. Each fixes a positive system, enumerates every π-system, closes it under root reflections, and recovers all 5,079 closed root subsystems, counting both the empty subsystem and the full root system Φ. Exactly 103 are maximal proper:

$$27D_5 + 36(A_5 + A_1) + 40A_2^3.$$

This verifies the three maximal-subsystem orbits used in Section 4. The homogeneous census in Section 3 is reconstructed separately in Appendix A.

Both E_6 implementations then evaluate all 38,880 triples and independently reconstruct the pairwise distributions, common intersections, six cells, and two extremal orbits. The primary implementation additionally reconstructs the arithmetic characterizations, stabilizers, and root-system containments; separate audit programs reconstruct the chiral-dimension table and global-kernel arithmetic. The deterministic certificate `e6_extrema_certificate.json` records the root numbering, reflection permutations, six orbit representatives, subsystem counts, and all finite distributions used in the proofs. The two implementations agree on their shared incidence surface; the primary implementation and separate audit programs supply the additional data reported below.

The chiral-dimension audit uses the minuscule **27**. Its weights pair with every coroot in $\{-1, 0, 1\}$; hence every irreducible constituent on a regular root subgroup is also minuscule. Each constituent has one Weyl orbit of weights, with multiplicity one. Distinct constituents have disjoint orbits, and negation exchanges dual orbits or preserves a self-dual orbit. Thus cancelling opposite weights computes the irreducible chiral dimension for these restrictions. The verifier checks this domain and rejects nonminuscule counterexamples to the unrestricted weight-cancellation rule.

B.1 Reconstructed data

Table 3: Finite objects reconstructed from the Cartan matrix.

quantity	value
number of roots $\lvert\Phi(E_6)\rvert$	72
highest-root coefficients (Bourbaki node order)	$(1,2,2,3,2,1)$
Weyl-group order $\lvert W(E_6)\rvert$	51,840
$\lvert\mathcal{O}_D\rvert$	27
$\lvert\mathcal{O}_A\rvert$	36
$\lvert\mathcal{O}_\Theta\rvert$	40
closed root subsystems	5,079
maximal proper closed subsystems	103
triples scanned $\lvert\mathcal{X}\rvert$	38,880

Table 4: Complete pairwise intersection counts.

pair	roots in the intersection	number of pairs
D_5 and A_5+A_1	20	432
D_5 and A_5+A_1	16	540
D_5 and A_2^3	10	1,080
A_5+A_1 and A_2^3	14	360
A_5+A_1 and A_2^3	6	1,080

For a triple (D, A, Θ), put

$$p(D, A, \Theta) = \big(|D \cap A|, |D \cap \Theta|, |A \cap \Theta|\big).$$

The following table gives the complete joint distribution of this profile and the size of the common intersection.

Table 5: All 38,880 triples.

profile $p(D,A,\Theta)$	common type	common roots	triples
$(20,10,14)$	A_2+A_1	8	4,320
$(20,10,6)$	A_1+A_1	4	12,960
$(16,10,14)$	$A_2+A_1+A_1$	10	2,160
$(16,10,14)$	$A_1+A_1+A_1$	6	3,240
$(16,10,6)$	$A_1+A_1+A_1$	6	3,240
$(16,10,6)$	A_1	2	12,960

The possible sizes of the common intersection are $2, 4, 6, 8, 10$. Their multiplicities are, in the same order,

$$12{,}960, \quad 12{,}960, \quad 6{,}480, \quad 4{,}320, \quad 2{,}160.$$

The pairwise-maximal triples and the triples with largest common intersection form Weyl orbits of sizes 4,320 and 2,160, with stabilizers of orders 12 and 24. The first set yields 720 distinct common root subsystems, each represented by six triples; the second yields 1,080, each represented by two. For each of the 1,080 pairs (D, Θ), the program finds four pairwise-maximal triples and two triples

with largest common intersection. Pairing any of the four with either of the two gives a root-system containment, for a total of 8,640 ordered pairs.

B.2 Release verification

The distributed Python verification suite is run from the source directory with

`make verify`.

The orchestrator executes the two independent root-system implementations, both complete closed-subsystem censuses, the context-normalizer, d_χ, global-kernel, toral-grading, exceptional rank-deficient, and classical highest-weight checks. It reconstructs the exceptional audit and the static certificate in temporary paths, compares both byte-for-byte, and verifies the certificate digest. A component is reported only after its optimization-independent runtime checks and comparisons have passed. The primary exhaustive implementation is reproduced below; the independent $\mathbb{R}^8$ implementation remains a separate source file.

The formal layer is run with `make lean`, the master notebook with `make notebook`, and the reproducible manuscript build with `make paper-release-check`. The command `make all` runs the complete local release gate; GitHub Actions runs the same four lanes in clean hosted environments.

The repository adds a formally checked layer for the finite incidence claims. The JSON root coordinates, simple-reflection permutations, and six orbit seeds generate a Lean 4 data module. Lean checks the permutations against the displayed E_6 Cartan matrix, reconstructs the three subsystem orbits, and checks that their masks are closed root subsystems with component root-count/rank signatures $(40, 5)$, $(30, 5) + (2, 1)$, and $(6, 2)^3$. Standard simply-laced classification identifies these as D_5, $A_5 + A_1$, and A_2^3. Lean reconstructs all 38,880 triples, verifies that the six cell orbits partition the triple space, and checks the two extrema and the two cells with a closed irreducible six-root, rank-two component; standard classification identifies that component as A_2. It also checks the 8,640 labelled containment incidences by `native_decide`. The expected cardinalities are stated in Lean rather than imported from JSON. The disclosed trust boundary includes the compiler and interpreter path used by `native_decide`; the repository records the exact axiom receipt and rejects project-local proof escapes. This finite checker does not formalize the analytic compact-Lie and representation-theoretic classification used in Sections 2–3, nor the physical interpretation in Section 5; those boundaries are recorded explicitly in the repository.

The SHA-256 hashes of the two exhaustive implementations are

```
verify_e6_extrema.py
585a1de39c4b031c4c285d14e96d04bd12ade519ae9b71482f33d6ccafc426eb
verify_e6_extrema_r8_fast.py
46f402c2bd9165d51e3fca48464fcf906ad685c9a1342fe4185131bba8e6041d
```

Listing 1: Exhaustive Python verification of the finite root-system assertions and counts underlying Theorems 4.2 and 4.3.

```python
#!/usr/bin/env python3
"""Exhaustively verify the two E6 intersection extrema and their classical incidence data."""

from collections import Counter, defaultdict, deque
from fractions import Fraction
from itertools import combinations, product

from verification_checks import require

# Bourbaki numbering: 1--3--4--5--6, with node 2 attached to 4.
C = (
    (2, 0, -1, 0, 0, 0),
```

```
    (0, 2, 0, -1, 0, 0),
    (-1, 0, 2, -1, 0, 0),
    (0, -1, -1, 2, -1, 0),
    (0, 0, 0, -1, 2, -1),
    (0, 0, 0, 0, -1, 2),
)
N = len(C)
E = tuple(tuple(int(i == j) for j in range(N)) for i in range(N))


def neg(v):
    return tuple(-x for x in v)


def add(u, v):
    return tuple(x + y for x, y in zip(u, v))


def pairing(weight, root):
    return sum(x * y for x, y in zip(weight, root))


def root_inner(u, v):
    return sum(u[i] * C[i][j] * v[j] for i in range(N) for j in range(N))


def reflect_root(root, i):
    out = list(root)
    out[i] -= sum(root[j] * C[j][i] for j in range(N))
    return tuple(out)


def reflect_weight(weight, i):
    return tuple(weight[j] - weight[i] * C[i][j] for j in range(N))


def reflect_set(S, i):
    return frozenset(reflect_root(a, i) for a in S)


def orbit(seed, step):
    seen, queue = {seed}, deque([seed])
    while queue:
        x = queue.popleft()
        for i in range(N):
            y = step(x, i)
            if y not in seen:
                seen.add(y)
                queue.append(y)
    return frozenset(seen)


def row_reduce(vectors):
    rows = [list(map(Fraction, v)) for v in vectors]
    r = 0
    for c in range(N):
        pivot = next((j for j in range(r, len(rows)) if rows[j][c]), None)
        if pivot is None:
            continue
        rows[r], rows[pivot] = rows[pivot], rows[r]
        q = rows[r][c]
        rows[r] = [x / q for x in rows[r]]
        for j, row in enumerate(rows):
            if j != r and row[c]:
                q = row[c]
                rows[j] = [x - q * y for x, y in zip(row, rows[r])]
        r += 1
    return rows, r


def rank(vectors):
    return row_reduce(vectors)[1]


def inverse(matrix):
    rows = [
        list(map(Fraction, row)) + [Fraction(i == j) for j in range(N)]
        for i, row in enumerate(matrix)
    ]
    for c in range(N):
        pivot = next(j for j in range(c, N) if rows[j][c])
        rows[c], rows[pivot] = rows[pivot], rows[c]
        q = rows[c][c]
```

```
        rows[c] = [x / q for x in rows[c]]
        for j in range(N):
            if j != c and rows[j][c]:
                q = rows[j][c]
                rows[j] = [x - q * y for x, y in zip(rows[j], rows[c])]
    return tuple(tuple(row[N:]) for row in rows)


C_INV = inverse(C)


def weight_inner(u, v):
    return sum(u[i] * C_INV[i][j] * v[j] for i in range(N) for j in range(N))


def components(S):
    remaining, answer = set(S), []
    while remaining:
        start = remaining.pop()
        component, queue = {start}, [start]
        while queue:
            a = queue.pop()
            adjacent = {b for b in remaining if root_inner(a, b)}
            remaining -= adjacent
            component |= adjacent
            queue.extend(adjacent)
        answer.append(frozenset(component))
    return tuple(sorted(answer, key=lambda x: (len(x), sorted(x))))


def root_type(S):
    size_to_type = {2: "A1", 6: "A2", 12: "A3", 20: "A4", 30: "A5", 40: "D5"}
    sizes = sorted((len(c) for c in components(S)), reverse=True)
    return "+".join(size_to_type[n] for n in sizes)


def closed(S, roots):
    return all(neg(a) in S for a in S) and all(
        add(a, b) not in roots or add(a, b) in S for a in S for b in S
    )


def check_subsystems(sets, roots, count, size, dimension):
    require(len(sets) == count)
    require({(len(S), rank(S), closed(S, roots)) for S in sets} == {
        (size, dimension, True)
    })


def main():
    roots = orbit(E[0], reflect_root)
    highest = max((a for a in roots if min(a) >= 0), key=sum)
    require(len(roots) == 72 and highest == (1, 2, 2, 3, 2, 1))
    require(all(root_inner(a, a) == 2 for a in roots))

    # Bit masks make the Weyl-orbit checks independent of Python's
    # object-heavy frozenset representation.
    root_list = sorted(roots)
    root_index = {root: i for i, root in enumerate(root_list)}
    reflection_permutations = tuple(
        tuple(root_index[reflect_root(root, i)] for root in root_list)
        for i in range(N)
    )

    def set_mask(S):
        return sum(1 << root_index[root] for root in S)

    def permute_mask(mask, permutation):
        out = 0
        while mask:
            bit = mask & -mask
            index = bit.bit_length() - 1
            out |= 1 << permutation[index]
            mask -= bit
        return out

    def tuple_mask_orbit(seed):
        seen, queue = {seed}, deque([seed])
        while queue:
            item = queue.popleft()
            for permutation in reflection_permutations:
                image = tuple(permute_mask(mask, permutation) for mask in item)
                if image not in seen:
```

```
                    seen.add(image)
                    queue.append(image)
        return frozenset(seen)

    # The E6 Weyl group order is used only in orbit-stabilizer divisions.
    weyl_order = 51_840

    weights = orbit(E[0], reflect_weight)  # Dynkin labels of W omega_1
    D = {w: frozenset(a for a in roots if pairing(w, a) == 0) for w in weights}
    root_pairs = {min(a, neg(a)) for a in roots}
    A = {
        a: frozenset(b for b in roots if b in (a, neg(a)) or root_inner(a, b) == 0)
        for a in root_pairs
    }
    theta0 = frozenset(a for a in roots if a[highest.index(3)] % 3 == 0)
    Thetas = orbit(theta0, reflect_set)
    D_sets, A_sets = frozenset(D.values()), frozenset(A.values())
    D_masks = {d: set_mask(d) for d in D_sets}
    A_masks = {a: set_mask(a) for a in A_sets}
    Theta_masks = {theta: set_mask(theta) for theta in Thetas}
    check_subsystems(D_sets, roots, 27, 40, 5)
    check_subsystems(A_sets, roots, 36, 32, 6)
    check_subsystems(Thetas, roots, 40, 18, 6)

    theta_factors = {}
    for theta in Thetas:
        factors = components(theta)
        require(tuple(map(len, factors)) == (6, 6, 6))
        theta_factors[theta] = {a: factor for factor in factors for a in factor}

    DA = Counter(
        (pairing(w, a) != 0, len(d & q))
        for (w, d), (a, q) in product(D.items(), A.items())
    )
    DT = Counter(len(d & t) for d, t in product(D_sets, Thetas))
    AT = Counter((a in t, len(q & t)) for (a, q), t in product(A.items(), Thetas))
    require(DA == Counter({(False, 16): 540, (True, 20): 432}))
    require(DT == Counter({10: 1_080}))
    require(AT == Counter({(False, 6): 1_080, (True, 14): 360}))
    pairwise_maxima = max(n for _, n in DA), max(DT), max(n for _, n in AT)

    profiles, common_sizes = Counter(), Counter()
    cells = defaultdict(set)
    cell_types = defaultdict(set)
    pairwise, total = set(), set()
    pairwise_by_dt, total_by_dt = defaultdict(list), defaultdict(list)
    pairwise_intersections, total_intersections = Counter(), Counter()

    for (w, d), (a, q), theta in product(D.items(), A.items(), Thetas):
        profile = len(d & q), len(d & theta), len(q & theta)
        common = d & q & theta
        profiles[profile] += 1
        common_sizes[(profile, len(common))] += 1

        is_pairwise = profile == pairwise_maxima
        factor = theta_factors[theta].get(a)
        nontrivial_on_factor = factor is not None and any(
            pairing(w, b) != 0 for b in factor
        )
        is_total = a in theta and pairing(w, a) == 0 and nontrivial_on_factor
        require(is_total == (len(common) == 10))

        triple = (D_masks[d], A_masks[q], Theta_masks[theta])
        cells[profile, len(common)].add(triple)
        cell_types[profile, len(common)].add(root_type(common))
        if is_pairwise:
            require((len(common), rank(common)) == (8, 3))
            pairwise.add(triple)
            pairwise_by_dt[d, theta].append((q, common))
            pairwise_intersections[common] += 1
        if is_total:
            require((len(common), rank(common)) == (10, 4))
            total.add(triple)
            total_by_dt[d, theta].append((q, common))
            total_intersections[common] += 1

    require(profiles == Counter({
        (20, 10, 14): 4_320,
        (20, 10, 6): 12_960,
        (16, 10, 14): 5_400,
        (16, 10, 6): 16_200,
    }))
    require(common_sizes == Counter({
```

```
        ((16, 10, 6), 2): 12_960,
        ((20, 10, 6), 4): 12_960,
        ((16, 10, 6), 6): 3_240,
        ((16, 10, 14), 6): 3_240,
        ((20, 10, 14), 8): 4_320,
        ((16, 10, 14), 10): 2_160,
    }))
    expected_types = {
        ((16, 10, 6), 2): {"A1"},
        ((20, 10, 6), 4): {"A1+A1"},
        ((16, 10, 6), 6): {"A1+A1+A1"},
        ((16, 10, 14), 6): {"A1+A1+A1"},
        ((20, 10, 14), 8): {"A2+A1"},
        ((16, 10, 14), 10): {"A2+A1+A1"},
    }
    require(dict(cell_types) == expected_types)
    require({key for key, types in cell_types.items() if any("A2" in t for t in types)} == {
        ((20, 10, 14), 8), ((16, 10, 14), 10)
    })

    require(len(pairwise) == 4_320 and len(total) == 2_160)

    require(len(cells) == 6)
    for triples in cells.values():
        require(tuple_mask_orbit(next(iter(triples))) == frozenset(triples))

    pairwise_orbit = tuple_mask_orbit(next(iter(pairwise)) )
    total_orbit = tuple_mask_orbit(next(iter(total)))
    require(pairwise_orbit == frozenset(pairwise))
    require(total_orbit == frozenset(total))
    require(weyl_order // len(pairwise_orbit) == 12)
    require(weyl_order // len(total_orbit) == 24)

    require(len(pairwise_intersections) == 720)
    require(Counter(pairwise_intersections.values()) == Counter({6: 720}))
    require(len(total_intersections) == 1_080)
    require(Counter(total_intersections.values()) == Counter({2: 1_080}))
    require(set(pairwise_by_dt) == set(total_by_dt) and len(pairwise_by_dt) == 1_080)
    require(Counter(map(len, pairwise_by_dt.values())) == Counter({4: 1_080}))
    require(Counter(map(len, total_by_dt.values())) == Counter({2: 1_080}))
    require(all(len({common for _, common in entries}) == 1
                for entries in total_by_dt.values()))

    labelled_containments, inclusions = 0, set()
    for key in pairwise_by_dt:
        d, theta = key
        for q_p, r_p in pairwise_by_dt[key]:
            for q_t, r_t in total_by_dt[key]:
                require(r_p < r_t)
                require((d & q_p) & (d & q_t) == r_p)
                require(rank((d & q_p) | (d & q_t)) == rank(d & q_t))
                labelled_containments += 1
                inclusions.add((set_mask(r_p), set_mask(r_t)))
    require(labelled_containments == 8_640 and len(inclusions) == 2_160)
    inclusion_orbit = tuple_mask_orbit(next(iter(inclusions)))
    require(inclusion_orbit == frozenset(inclusions))

    # Classical cubic-surface incidence carried by the 27 weights.
    require({weight_inner(w, w) for w in weights} == {Fraction(4, 3)})
    require(Counter(
        weight_inner(u, v) for u, v in combinations(weights, 2)
    ) == Counter({Fraction(1, 3): 216, Fraction(-2, 3): 135}))

    tritangents = {
        frozenset((u, v, neg(add(u, v))))
        for u, v in combinations(weights, 2)
        if neg(add(u, v)) in weights and len({u, v, neg(add(u, v))}) == 3
    }
    require(len(tritangents) == 45)
    require(orbit(
        next(iter(tritangents)),
        lambda T, i: frozenset(reflect_weight(w, i) for w in T),
    ) == frozenset(tritangents))

    double_sixes = set()
    for a in root_pairs:
        plus = {w for w in weights if pairing(w, a) > 0}
        minus = {w for w in weights if pairing(w, a) < 0}
        require((len(plus), len(minus)) == (6, 6))
        require({weight_inner(u, v) for u, v in combinations(plus, 2)} == {
            Fraction(1, 3)
        })
        require({weight_inner(u, v) for u, v in combinations(minus, 2)} == {
```

```
            Fraction(1, 3)
        })
        require(Counter(weight_inner(u, v) for u in plus for v in minus) == Counter({
            Fraction(-2, 3): 30,
            Fraction(1, 3): 6,
        }))
        double_sixes.add(frozenset(plus | minus))
    require(len(double_sixes) == 36)

    nine_line_sets, partitions = set(), set()
    for theta in Thetas:
        blocks = frozenset(
            frozenset(w for w in weights if all(pairing(w, a) == 0 for a in factor))
            for factor in components(theta)
        )
        require(sorted(map(len, blocks)) == [9, 9, 9])
        require(len(set().union(*blocks)) == 27)
        partitions.add(blocks)
        nine_line_sets |= set(blocks)
    require(len(partitions) == 40 and len(nine_line_sets) == 120)
    require(all(sum(T <= block for T in tritangents) == 6 for block in nine_line_sets))

    print("E6 extremal-incidence verification passed")
    print(f"roots={len(roots)}, |W|={weyl_order:,}, subsystem orbits=(27, 36, 40)")
    print(f"triples={sum(profiles.values()):,}")
    print(f"pairwise maximizers={len(pairwise):,}, stabilizer={weyl_order//len(pairwise):,}")
    print(f"common maximizers={len(total):,}, stabilizer={weyl_order//len(total):,}")
    print(f"joint-profile orbits={len(cells)}")
    for key in sorted(common_sizes):
        t = next(iter(cell_types[key]))
        print(f"profile={key[0]}, common roots={key[1]}, type={t}: {common_sizes[key]:,}")
    print("A2 occurs exactly in the pairwise- and common-maximal cells")
    print(f"distinct inclusions={len(inclusions):,}; labelled realizations={labelled_containments:,}")
    print("cubic-surface data: 27 lines, 36 double-sixes, 45 tritangents, 40 triads")


if __name__ == "__main__":
    main()
```

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